Light Collection in a Polished Plastic Scintillator Read Out by Side-Mounted SiPMs: a Geant4 Matrix Simulation Study

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Abstract

We describe a Geant4 (v11.3.2) simulation of a 250 mm×250 mm×10 mm polished plastic scintillator (PVT, UPS-92S/EJ-200 class) read out by side-coupled silicon photomultipliers (SiPMs), modelled on the muon-trigger detector of Rao *et al.* [J. Appl. Phys. **138**, 114504 (2025)]. The detector is exposed to a vertical 4 GeV muon pencil beam. We document the geometry and the optical-physics implementation in detail, and state the actual input data rather than blanket assumptions: the Geant4 *unified* surface model and the finish/type combinations that represent the four wrapping configurations (bare plate in a black holder, TiO_2 paint, aluminium foil, high-gloss paper); the digitized scintillator emission and bulk-absorption curves; the digitized Kuraray Y-11 WLS absorption and re-emission spectra used by the config-4 fibre core (with absorption length 1.0 mm at 430 nm); the digitized ASD-NUV4S-P-40 photocathode PDE that drives photodetection; and the per-event optical accounting that feeds the analysis. Config 4 is built from an analytic, multi-clad Y-11 fibre bundle in a white Lambertian clamp with exponential (7 ns) WLS re-emission. Two calibration scans fix the remaining surface inputs: a 99-file bare-contact scan (11 fractions, 99 000 events) selects an effective plate–holder contact fraction $f = 0.045$, and an 18-file aluminium-backing scan (18 000 events; the backing model is unchanged from the earlier production campaign) selects air-backed foil. With these parameters fixed, a 48-file matrix scan (4 SiPM layouts × 4 reflectors × 3 beam positions, 2500 events each, 120 000 events) and a whole-plate irradiation (25 000 events per configuration and wrapper, 400 000 events) are produced, for 637 000 events in total. With these calibrated surfaces the three direct-coupled layouts recover the measured ordering bare < TiO_2 < Al < glossy: glossy paper is brightest, air-backed Al reaches about 92 % of the glossy yield, and TiO_2 is the most position-dependent wrapper. The WLS-fibre layout retains a different ordering, TiO_2 < bare < Al < glossy, and remains roughly half as efficient as config 3 for the reflective wrappers. The whole-plate contrast ranks TiO_2 highest ($\kappa_{2D} = 1.91$–8.31), followed by the contact-limited bare plate (1.36–3.82), with Al (1.08–1.28) and glossy paper (1.07–1.28) nearly uniform. The clean air-gap reference ($f = 0$) is retained to expose the contact systematic. We report per-event photon yields, photoelectron spectra, wavelength distributions, position-resolved efficiency maps, and the contact/aluminium systematics. Model and implementation caveats are documented, principally the residual config-4/config-3 efficiency ratio.

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1 Introduction and motivation

Plastic scintillators read out at their edges by SiPMs are attractive for large-area, low-cost muon triggers and veto systems for cosmic-ray muon tomography [18, 19]. The amount of scintillation light that reaches the photosensor is governed almost entirely by how the non-instrumented faces of the plate are treated: a polished plate in air traps light by total internal reflection (TIR) and can be a nearly lossless light guide, whereas a plate wrapped in an absorbing or diffusive material loses photons at every bounce. Understanding this trade-off quantitatively is the central design question, and it is precisely where simulation is most useful, because the photon transport is a multi-bounce random walk that is difficult to model analytically [7].

This report documents a Geant4 (v11.3.2) [2, 3] simulation built to address that question for the specific geometry and surface treatments of Rao *et al.* [1]. The deliverable is a self-contained "matrix scan": every combination of four SiPM layouts and four reflector treatments, at three beam positions on the plate. The goal is not to fit a single number but to test how far the simulation reproduces the *ordering* of light-collection efficiency that the experiment observes, and to expose the physics behind both the parts it reproduces and the part it does not.

The remainder of this document is organised as follows. Section 2 describes the detector model: the scintillator, the SiPM read-out, the four read-out layouts (including the wavelength-shifting fibre clamp of config 4), the reflector treatments, the tabulated optical inputs that the transport actually consumes (emission, bulk and WLS attenuation, refractive index, reflector reflectivity, wavelength-shifting and photocathode PDE), and the optical-physics implementation, all verified

line-by-line against the Geant4 source. Section 3 describes the simulation campaigns and fixes the two unmeasured surface inputs first: the beam and patch setup, the bare-holder contact scan, and the aluminium grease-versus-air comparison. These select the nominal contact fraction and Al backing used everywhere below. Section 4 then presents the matrix irradiation results obtained with those parameters—collection efficiencies, photon/photoelectron spectra, wavelengths, and position-resolved maps. Section 5 presents the complementary whole-slab irradiation, which samples the full plate uniformly. Section 6 documents known caveats, and Sec. 7 summarises.

2 Detector model

2.1 Scintillator plate

The active volume is a rectangular PVT scintillator $250 \times 250 \times 10$ mm, mirroring the UPS-92S plate of Ref. [1]. It is assigned the Geant4 NIST base material `G4_PLASTIC_SC_VINYLTOLUENE` with a custom optical properties table built from the EJ-200 and UPS-923A data sheets [11, 12, 13]. The optical constants—the refractive index ($n = 1.58$, essentially flat across the visible), light yield, emission peak, bulk attenuation length, scintillation decay and rise times, and the Birks constant [14, 15]—are collected in the consolidated parameter table of Sec. 2.5 and are not repeated here. Two entries are deliberate modelling choices rather than quoted numbers: the bulk attenuation length is peaked at 3.8 m but falls steeply in the UV to model wavelength-shifter self-absorption, and the emission spectrum (Fig. 4) is essentially zero below 380 nm and above 550 nm. Both scintillation and Čerenkov photons are produced and tracked (Sec. 2.7); the optical grid spans 310 nm to 700 nm so that the Čerenkov continuum, which extends well below the scintillation band, is covered.

2.2 SiPM readout and coupling

Each SiPM [8] is modelled mechanically as a $4 \times 4 \times 0.6$ mm box of "SiPM epoxy" ($n = 1.55$), coupled to a side face of the plate through a 0.1 mm silicone grease pad ($n = 1.46$). The coupling chain is therefore *scintillator* → *grease* → *SiPM*, with Fresnel transmission at each dielectric interface and photodetection at the final grease/SiPM boundary.

Four layouts are studied (schematic in Fig. 1):

Config 1 (8 SiPMs) two SiPMs per side face, at ±50% of the half-length.

Config 2 (4 SiPMs) one SiPM per side face, at the *same* ±50% offsets as config 1—i.e. config 1 with one device removed per side, the removed slot rotated from side to side so the layout remains four-fold symmetric.

Config 3 (2 SiPMs) both SiPMs on the *same* narrow face—the $-x$ face in the simulation frame, which is the report plan view's top edge (Sec. 3)—at ±50%, matching T1/T2 of Ref. [1].

Config 4 (1 SiPM + WLS clamp) a single SiPM reading out a wavelength-shifting (WLS) fibre clamp along the $-x$ narrow face (Sec. 2.4).

SiPM footprints are clamped so they cannot overhang the slab face.

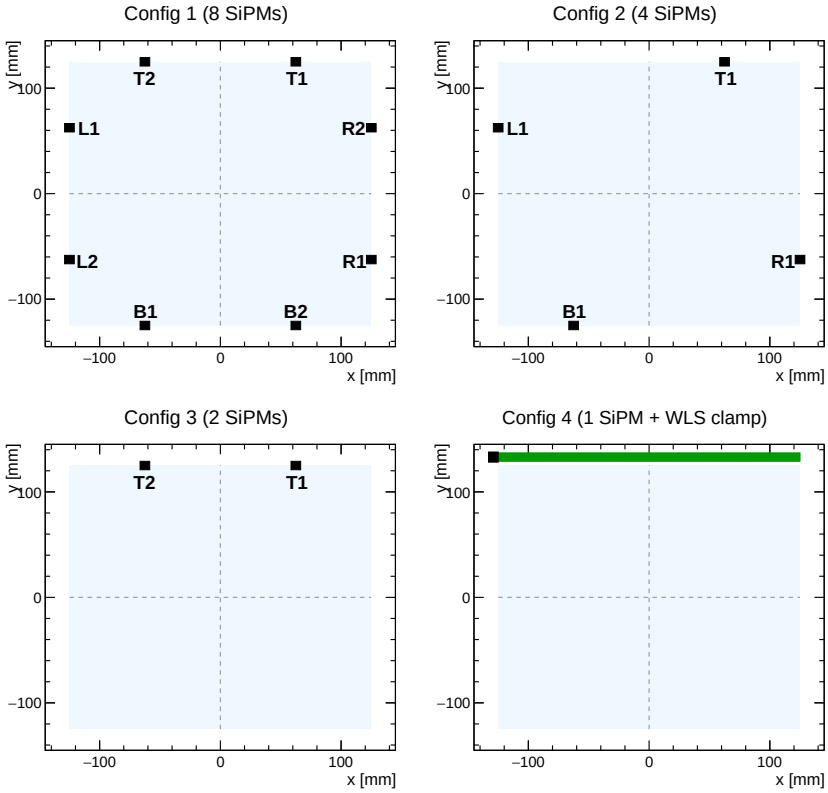


Figure 1: Schematic plan view of the four detector layouts on the 250 mm×250 mm plate footprint. Black squares mark the side-mounted SiPMs (drawn oversized for visibility; the active area is 4 mm×4 mm), and the dashed grey lines are the $x = 0$, $y = 0$ axes. Config 1 places two SiPMs on each of the four edges; config 2 places one SiPM per edge at the same offsets as config 1 (one device removed per side); config 3 places both SiPMs on the top ($+y$, report frame) edge; config 4 uses a single SiPM reading out a WLS fibre clamp on the same top edge, with the read-out SiPM at its left end (green bar), following Fig. 3(b) of Ref. [1]. In the simulation frame both configs 3 and 4 sit on the $-x$ narrow face (the report plan view maps $(x, y) \to (y, -x)$). The inwards-offset labels name the eight side-mounted sensors (T/B/R/L = top/bottom/right/left); on each edge "1" marks the sensor nearer the corner reached when travelling clockwise around the perimeter, so "1" sits at the top-right (T1), bottom-right (R1), bottom-left (B1) and top-left (L1) corners. This is the scheme used in the light-sharing discussion below. The same black-square markers are overlaid on the collection-efficiency maps (Figs. 20–23).

2.3 Reflector treatments

Four treatments are applied as a *logical skin* over the non-instrumented surfaces of the plate; the grease/SiPM patches override the skin locally with dedicated *border* surfaces. The mapping from each physical treatment to the Geant4 optical-surface parameters is the heart of the model—it determines the light-collection ordering—so it is given in full. Table 1 lists every property that is actually attached to each surface, and the subsubsections below describe each reflector physically, state how it is modelled, and assess how faithfully the model represents the real object. (A fifth finish, PTFE tape, exists in the code for completeness but is not part of the reference study.) The strong sensitivity of light collection to the wrapping and reflector choice is well established experimentally [21].

Table 1: Geant4 optical-surface parameters for the four reflectors, exactly as attached in `src/GeometryMuonScint.cc`. $R(\lambda)$ is tabulated on the shared ten-point 360 nm to 650 nm grid and quoted here as its range; the full arrays are in the code. A dash marks a property that is neither set nor used for that finish.

	Bare (black holder)	**TiO_2** (EJ-510 paint)	**Al foil (grease/air)** (backing choice)	**Glossy** (photo paper)
Physical contact	1.2 mm air gap	direct (optical)	grease film / air gap	air gap
Geant4 surface object	none*	skin	skin	skin
Type	`dielectric_dielectric`	`dielectric_dielectric`	`dielectric_dielectric`	`dielectric_dielectric`
Model	`glisur`*	`unified`	`unified`	`unified`
Finish	`polished`*	`groundfront-painted`	`polished-backpainted`	`polished-backpainted`
`sigma_alpha`	—	0.10 rad	0	0
`REFLECTIVITY` $R(\lambda)$	—	0.955–0.983	0.904–0.925	0.900–0.955
Backing `RINDEX`	—	—	1.46 grease / 1.0003 air	1.0003 (air)
`SPECULARSPIKE` f_{ss}	—	0	1	1
`SPECULARLOBE` f_{sl}	—	0	0	0
`BACKSCATTER` f_{bs}	—	0	0	0
Total internal reflection	yes (Fresnel/TIR)	no (Lambertian)	grease/air ($\leq 67.5°$ grease) (full cone air)	yes (full cone)

*The bare plate carries *no* surface object: Geant4 then applies its built-in `glisur/polished dielectric_dielectric` default, with Fresnel and TIR taken from the material refractive indices. The black holder shell, in contrast, does carry an absorbing skin (`MuonScintHolderAbsorber`, a `polished dielectric_dielectric` surface with `REFLECTIVITY` = `TRANSMITTANCE` = 0), so the sub-critical light that refracts across the air gap is killed at the wall—and the holder bulk, with its 1 µm absorption length, would absorb it anyway. The bare model also carries an optional mean-field contact fraction f (production default $f = 0.045$; the clean air-gap reference is $f = 0$); for $f > 0$ every hit on a bare face is absorbed with probability f ($R = 1 - f$, $T = 0$), representing partial plate–holder contact.

2.3.1 Bare plate in its black holder

Physical object. The polished UPS-92S plate is clamped into a black 3D-printed ABS enclosure, its faces simply resting near the wall.

Model. No wrapping skin is attached. The plate keeps its native polished scintillator ($n = 1.58$) against air ($n = 1.0003$) across a real 1.2 mm gap; light below the critical angle ($\arcsin(1/1.58) = 39.3°$) refracts across the gap and is absorbed in the holder, whose black ABS has an attenuation length of 1 µm, while light above it is returned by total internal reflection. Fresnel reflection and TIR are taken from the material `RINDEX` tables.

Fidelity. This is the least faithful configuration and the source of the report's central modelling choice. A real clamped plate touches the holder at scattered high spots, and each contact forms a frustrated-TIR sink that removes trapped light; surface scratches do the same. Those loss channels are absent in the clean air-gap model ($f = 0$), where the plate acts as a near-lossless light guide and is one of the *brightest* collectors, whereas the reference measurement gives it the *lowest* signal. The code therefore offers an optional mean-field contact fraction, `/dcs-monitor/det/muonScint/setBareContactFraction <f>` ($0 \leq f \leq 1$): for $f > 0$ every hit on a bare face is absorbed with probability f (a polished `dielectric_dielectric` skin with `REFLECTIVITY` $= 1 - f$ and `TRANSMITTANCE` $= 0$), representing frustrated TIR over an effective contact area. The loss is ignored for the config-4 bare layout, whose clamp occupies that face. The dedicated scan (Sec. 3.2) gives a nominal $f = 0.045$, the smallest sampled fraction that places all nine config/position cells below the corresponding TiO_2 reference; $f = 0$ is kept as the clean-air reference and lower bound on the loss.

2.3.2 Titanium-dioxide paint (EJ-510)

Physical object. A white, diffuse-reflectance paint applied directly to the plate (three coats, ≈ 0.11 mm), in optical contact with the scintillator.

Model. `groundfrontpainted` with a purely Lambertian reflection: a random draw against `REFLECTIVITY` returns the photon with a cosine-law distribution and *no* specular or TIR component ($f_{ss} = f_{sl} = f_{bs} = 0$). The reflectance rises from 0.955 in the UV to a plateau of 0.983 near 470 nm.

Fidelity. The Lambertian/no-TIR treatment is the physically correct description of light emerging from a thick, strongly scattering paint, and is this wrapper's strongest feature. The *absolute* reflectance curve is its weakest: no public spectral-reflectance data exist for EJ-510, so $R(\lambda)$ is a fitted representative white-paint curve (≈ 0.96–0.98), not a measurement of the actual coating, and no angular dependence of the diffuse lobe is modelled. The `sigma_alpha` value is set for bookkeeping but is inert at this finish (`groundfrontpainted` reflects diffusely regardless of the facet normal).

2.3.3 Aluminium foil

Physical object. Commercial shiny aluminium foil wrapped around the plate; per Ref. [1] optical grease is applied in places so that the foil contacts the scintillator rather than leaving an air gap.

Model. `polishedbackpainted` with a selectable backing refractive index, chosen pre-init with `/dcs-monitor/det/muonScint/setAlBacking air|grease`. The production choice is `air` ($n = 1.0003$), which leaves the full TIR cone intact and lets only the sub-critical fraction cross to the foil, modelling a foil separated from the plate by a trapped air layer. The alternative, `grease` ($n = 1.46$), represents a foil applied over the coupling grease, as in Ref. [1]; the scintillator/grease interface is a normal Fresnel/TIR boundary, but since the index step is small the critical angle is wide ($\arcsin(1.46/1.58) = 67.5°$): most light refracts into the grease film and is returned by the foil as a pure *specular* reflection ($f_{ss} = 1$). It is retained as the lower-bound systematic. In both cases `REFLECTIVITY` is the normal-incidence reflectance of evaporated aluminium computed from the McPeak *et al.* n, k data [20] (0.904 at 650 nm rising to 0.925 at 360 nm).

Fidelity. Two idealisations. (i) The tabulated reflectance is that of a smooth evaporated film and is therefore an *upper bound*: household foil is rougher and carries a native oxide, both of which lower R and add a small diffuse component we do not model. (ii) The foil is assumed to cover the plate uniformly, whereas real wrapping leaves wrinkles and trapped air. The production campaign uses the `air` backing, which retains the full TIR cone and recovers the measured near-equality of aluminium and glossy paper, reaching $\approx 92\,\%$ of the glossy yield in the direct-coupled layouts. The `grease` backing is retained as the lower-bound systematic (Sec. 3.3).

2.3.4 High-gloss photo paper

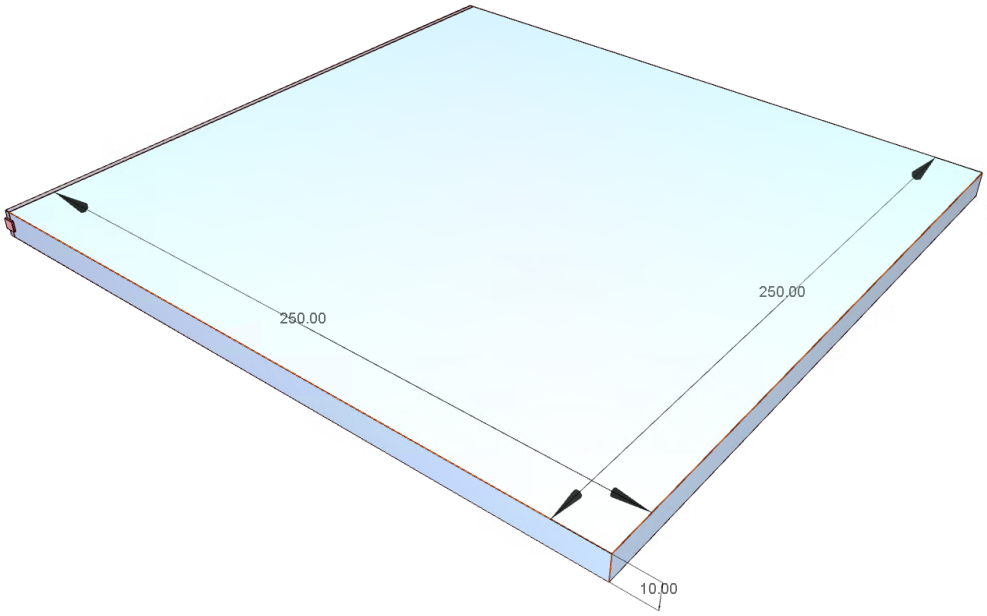
Physical object. A commercial super-high-gloss photographic paper (Print4Life, $300\,\mathrm{g\,m^{-2}}$) wrapped around the plate with an air gap (no grease).

Model. `polishedbackpainted` with the backing index set to air ($n = 1.0003$), so the plate retains its full TIR cone and only the sub-critical fraction crosses the gap to the paper, which returns it as a specular reflection ($f_{ss} = 1$) with a fitted $R(\lambda) \approx 0.900$–0.955—slightly lossier than the aluminium.

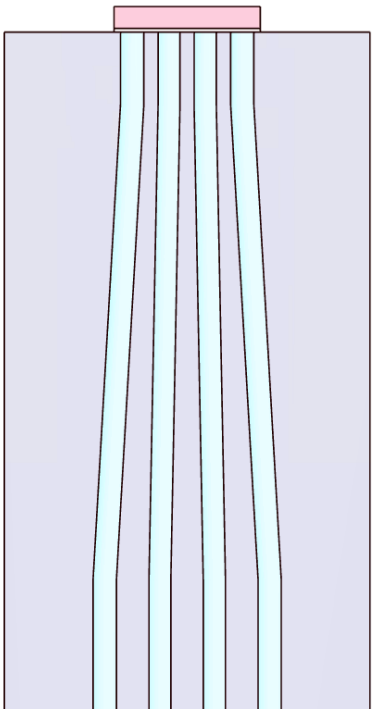
Fidelity. The reference reports glossy paper and aluminium foil as the two best wrappers (large, nearly equal signals), which motivates modelling glossy paper as a *slightly lossy mirror* rather than a diffuser. This is a model choice rather than a measurement: real photo paper is not a true mirror and has a diffuse component, and its $R(\lambda)$ is again fitted (no measured data), so this is the reflector whose optical model is least constrained by independent data.

2.4 Config 4: wavelength-shifting fibre clamp

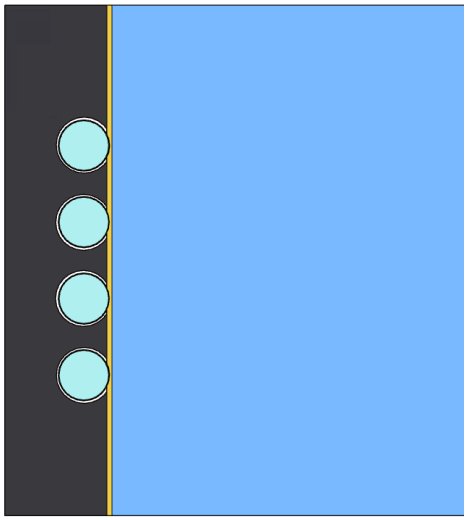
The fourth layout follows the single-SiPM, WLS-readout variant of Ref. [1]. Four 1.0 mm-diameter Kuraray Y-11(200) multi-clad wavelength-shifting fibres [16, 17] run the full 250 mm length of the $-x$ narrow face in a single row. Over most of the length they are parallel at a 2.5 mm centre-to-centre pitch—so a solid web of clamp material separates the milled 1.1 mm channels—and converge over a 41 mm length (39 mm ramp plus a 2 mm stub) to a touching 1.0 mm bundle that presents all four ends to a single 4 mm×4 mm SiPM at the $-y$ end; their far $+y$ ends are closed by individual aluminium mirror caps. Each fibre is modelled as a multi-clad guide: a Y-11 core ($n = 1.59$) inside a PMMA inner cladding ($n = 1.49$) inside a low-index fluoropolymer outer sheath ($n = 1.42$)—the low-index staircase that sustains core–cladding total internal reflection. In the production geometry the fibre solid is an analytic, mitered set of `G4CutTubs` legs (one per straight segment of the bent centreline), placed so adjoining legs meet at exact miter planes; this facet-free construction is what the runs use. (An earlier tessellated swept cylinder, which occasionally produced stuck optical tracks, is retained as an optional visualisation-only variant.) A thin 0.10 mm grease film couples the plate face to the clamp, and each fibre protrudes 0.05 mm through its channel mouth into that film, keeping the grease continuous against the scintillator. The deliberate 0.05 mm air annulus between the 1.0 mm fibre and its 1.1 mm channel is essential: the fibre must not touch the clamp ($n = 1.55$), and the outer sheath ($n = 1.42$) needs the low-index clearance to keep guiding the light. The clamp body is 3D-printed white plastic, modelled by default as a Lambertian reflector with 85 % reflectivity applied to *every* clamp face (channel walls and grease-facing front included), so light leaking out of the fibre is returned rather than lost. The far-end mirror is a full aluminium disc whose logical volume carries a reflecting skin as well as the fibre–cap border surfaces, so that light travelling in the *core*—not only the cladding—is also turned around at the far end. WLS re-emission is drawn from the digitized Y-11 emission spectrum with an exponential delay of 7 ns (Sec. 2.10), and the `G4OpWLS` process is enabled automatically when this geometry is built. In the report's top-down frame the clamp runs along the *top* edge with the single read-out SiPM at the *top-left* corner, as in Fig. 3(b) of Ref. [1].



(a) Three-dimensional view of the complete config-4 assembly, with the principal dimensions annotated.

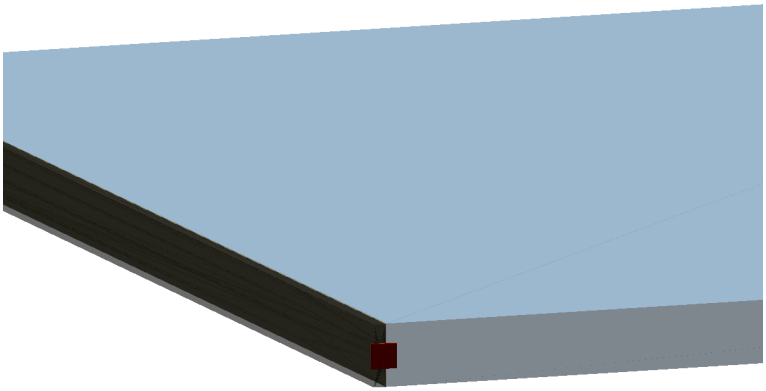


(b) Top view of the black fibre clamp: the four fibre channels and the read-out SiPM at the end of the converged bundle (report-frame top-left).

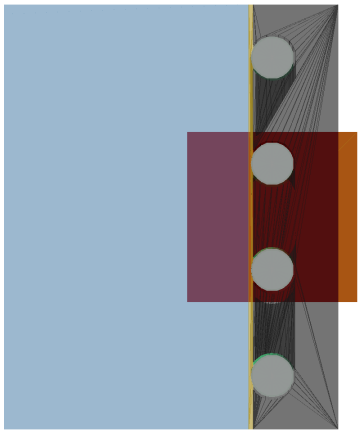


(c) Cross-section through the clamp: the scintillator, the thin grease film in the slab–clamp gap, the four fibre discs seated in their 1.1 mm channels with a 0.05 mm air annulus, and the outboard black clamp.

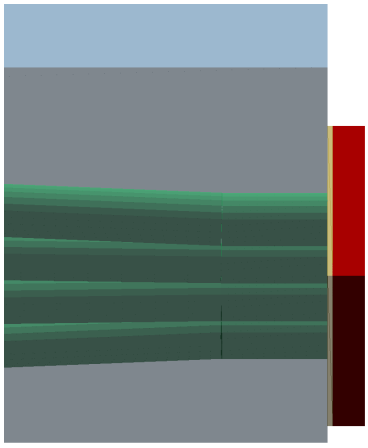
Figure 2: Geometry-only views of the config-4 WLS-fibre clamp, rendered from the Geant4 VRML geometry export (*no* beam-on). Following the event-display colour scheme, fibre cores are pale green, cladding dark green, the grease film amber, the black clamp and holder dark grey, and the read-out SiPM red. The four 1.0 mm fibres sit on the $-x$ narrow face in 1.1 mm channels separated by clamp webs, with only a thin grease film in the slab–clamp gap; they converge to a bundle read out at $-y$ by one SiPM and mirrored at $+y$. These screenshots show the clamp in its earlier black visualisation colour; the production clamp is white (Fig. 3).



(a) Oblique view of the whole plate with the clamp strip on the $-x$ narrow face and the single read-out SiPM at the $-y$ end; the fibres run the full plate length in y.



(b) Cross-section through the clamp: scintillator, grease film, and the four multi-clad fibre discs seated in their channels.



(c) Close-up of the four fibre ends converging onto the single read-out SiPM (here the Boolean clamp and grease film are hidden).

Figure 3: OpenGL (`TSG_OFFSCREEN`) geometry renders of the config-4 assembly (multi-clad $n = 1.59/1.49/1.42$ fibres, white clamp), generated head-lessly from Geant4 (no beam-on, drawn without wireframe edges). Panels (a) and (b) use the *tessellated* visualisation preset (`macros/muon_vis_wls_ogl.mac`): it is geometrically equivalent to the production analytic preset—identical round 1.0 mm fibres and clamp/grease volumes—but its clamp and grease film render as solids, whereas the analytic cut-tube cutters do not implement `CreatePolyhedron()` and would appear as point clouds. Panel (c) instead uses the *analytic production* geometry (`macros/muon_vis_wls_ogl_coupling.mac`) with only those two Boolean solids hidden, so that the plain `G4CutTubs` fibres, the read-out pad and the SiPM are seen coupling. These renders complement the hand-made VRML screenshots of Fig. 2.

2.5 Summary of detector parameters

The principal parameters of the three active components—the scintillator plate, the WLS fibre, and the SiPM—are collected in Table 2, alongside the corresponding values quoted for the reference experiment of Ref. [1]. Where a quantity is not reported in Ref. [1] or is not modelled here, a dash is shown; the simulation follows the experimental values wherever they are available, and the optical model of Sec. 2.7 is built entirely from the values in the last column.

Table 2: Principal parameters of the scintillator, the WLS fibre, and the SiPM, comparing the reference experiment of Ref. [1] with the values adopted in this simulation.

Parameter	Ref. [1]	This simulation
Scintillator (UPS-92S / PVT)		
Material	UPS-92S (AdvTECH UK)	PVT (`G4_PLASTIC_SC_VINYLTOLUENE`)
Dimensions	$250 \times 250 \times 10$ mm	$250 \times 250 \times 10$ mm
Refractive index	1.58	1.58
Density	1.05 g cm^{-3}	1.032 g cm^{-3}
Light output	50–60% of anthracene (≈ 8 to 10×10^3 ph/MeV)	1.0×10^4 MeV^{-1}
Emission peak	425 nm	425 nm (2.917 eV)
Bulk attenuation length	2000 mm	3.8 m (peak, falling in the UV)
Scintillation decay time	2.4 ns	2.1 ns
Rise time	—	0.9 ns
Birks constant	—	0.126 mm MeV^{-1}
Surface finish	all sides polished	all sides polished
WLS fibre (Kuraray Y-11(200))		
Fibre diameter	1.0 mm	1.0 mm (core+clad)
Cladding	multi-cladding	multi-clad: PMMA ($n = 1.49$) + fluoropolymer sheath ($n = 1.42$)
Core refractive index	—	1.59
Absorption peak	430 nm	430 nm (digitized Kuraray Y-11)
Emission peak	476 nm	476 nm
Attenuation length	> 3500 mm	$\gtrsim 5$ m (transport)
WLS absorption length at 430 nm	—	1.0 mm (systematic 0.3 mm to 3 mm)
WLS decay time	—	7 ns (exponential)
Read-out arrangement	single row, 250 mm long	4 fibres, 2.5 mm pitch $\rightarrow$ 1.0 mm bundle (41 mm convergence)
Channel diameter	—	1.1 mm (0.05 mm air annulus)
SiPM (CEFLA S.C. ASD-NUV4S-P-40)		
Active area	4×4 mm^2	4×4 mm^2
Number of cells	9340	—
Peak sensitivity	420 nm	410 nm
Photon detection efficiency	43%	0.431 (peak, digitized curve)
Epoxy refractive index	1.5115	1.55
Optical coupling	BC-630 grease	0.1 mm grease film ($n = 1.46$)
Gain / dark-count rate / capacitance	10^6 / 50–100 kHz/mm^2 / 0.84 nF	— (light transport only)

2.6 Optical property tables

Figure 4 plots the spectral input tables actually fed to Geant4, over the 310 nm to 700 nm range used in the transport. The left panel compares the normalised spectral shapes: the scintillator emission peaks at 425 nm, the Y-11 absorption band at 430 nm, and the Y-11 re-emission lies in the green at 476 nm—the shift that makes the config-4 read-out spectrum green. Drawing the WLS absorption and emission on one axis makes their overlap explicit, while the absolute WLS absorption strength is carried by the middle panel. That panel gives the attenuation lengths on a logarithmic scale: the scintillator is transparent across the visible but its bulk attenuation

collapses in the UV, whereas the digitized Y-11 WLS absorption is effectively transparent redward of 500 nm and rises to its 430 nm peak (absorption length 1 mm); the WLS absorption onset is marked by the dashed line near 486 nm. The right panel is the SiPM photocathode efficiency = PDE(λ), the digitized ASD-NUV4S-P-40 curve peaking at 0.431 near 410 nm. These curves are exactly the tabulated points of `src/GeometryMuonScint.cc`; Geant4 interpolates them linearly, so the drawn segments are what the simulation uses. The left and middle panels together therefore carry the complete WLS spectral input (emission and absorption), and no separate WLS-spectrum figure is needed.

The remaining input data are shown in Figs. 5–7, all plotted directly from the same source arrays: the time structure of the two optical sources—the scintillation pulse on linear and logarithmic scales, and the exponential Y-11 re-emission delay (Fig. 5); the tabulated reflectivity $R(\lambda)$ of the three reflective wrappers, also read out at the scintillation and WLS emission peaks (Fig. 6); and the refractive-index ladder, the multi-clad fibre's radial index staircase, and the critical angles implied by it (Fig. 7).

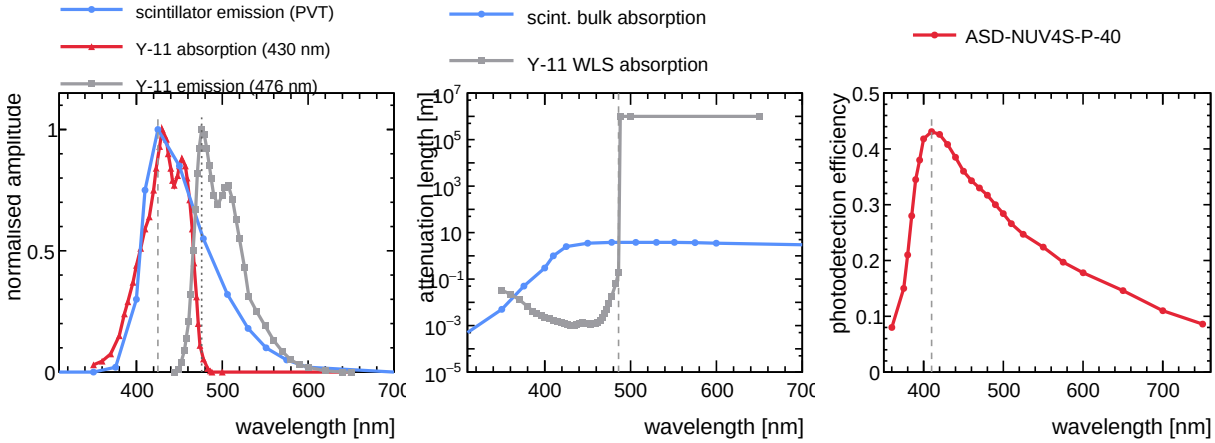


Figure 4: Optical property tables used by the simulation. *Left:* normalised spectral shapes of the PVT scintillator emission (peaking at 425 nm) and of the digitized Y-11 WLS fibre absorption (430 nm) and re-emission (476 nm); the absorption/emission overlap is the input to the `G4OpWLS` energy constraint $E'_\gamma \leq E_\gamma$. *Middle:* attenuation lengths on a logarithmic scale—the scintillator bulk absorption and the fibre WLS absorption (absolute scale set to 1 mm at the 430 nm peak). *Right:* SiPM photocathode PDE(λ), the digitized curve peaking at 410 nm. The vertical dashed line in each panel marks the characteristic wavelength of that panel's process—the scintillator emission peak at 425 nm and the Y-11 emission peak at 476 nm (left), the WLS absorption onset at 486 nm (middle), and the SiPM PDE peak at 410 nm (right); the scintillator emission peak and the SiPM PDE peak therefore nearly coincide, whereas the WLS absorption onset and the Y-11 emission peak are red-shifted into the WLS band. Markers are the tabulated points; segments are the linear interpolation Geant4 applies.

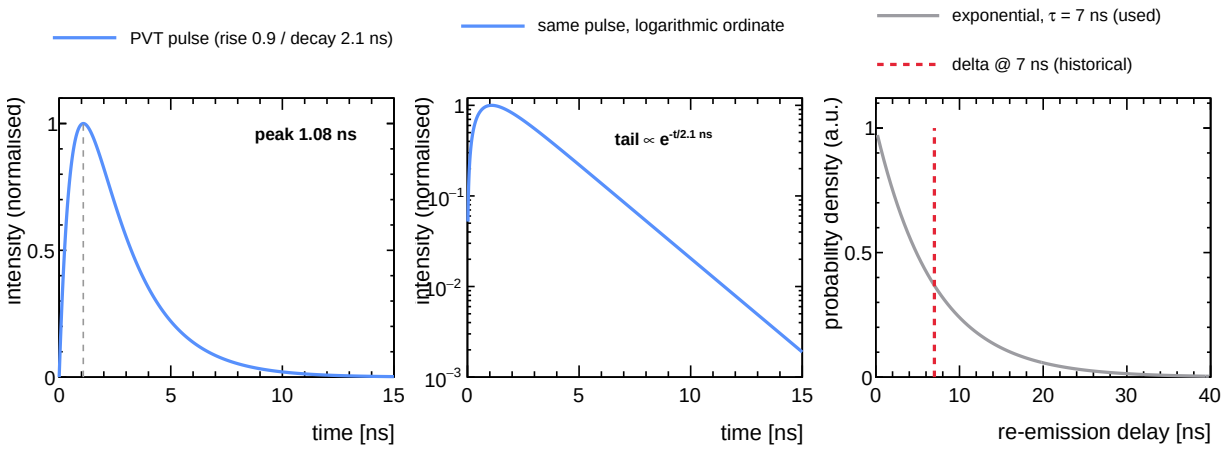


Figure 5: Time structure of the two optical sources. *Left:* the PVT scintillation pulse $f(t) \propto \left(1 - e^{-t/\tau_r}\right) e^{-t/\tau_d}$ with rise $\tau_r = 0.9\,\text{ns}$ and decay $\tau_d = 2.1\,\text{ns}$, normalised to its peak and shown on a linear scale; the dashed line marks the maximum at 1.08 ns. *Middle:* the same pulse on a logarithmic ordinate, exposing the exponential tail. *Right:* the Y-11 WLS re-emission probability density: exponential with $\tau = 7\,\text{ns}$ used in production (solid), compared with the historical fixed-delay profile at 7 ns (dashed). The corresponding spectral tables are in Fig. 4.

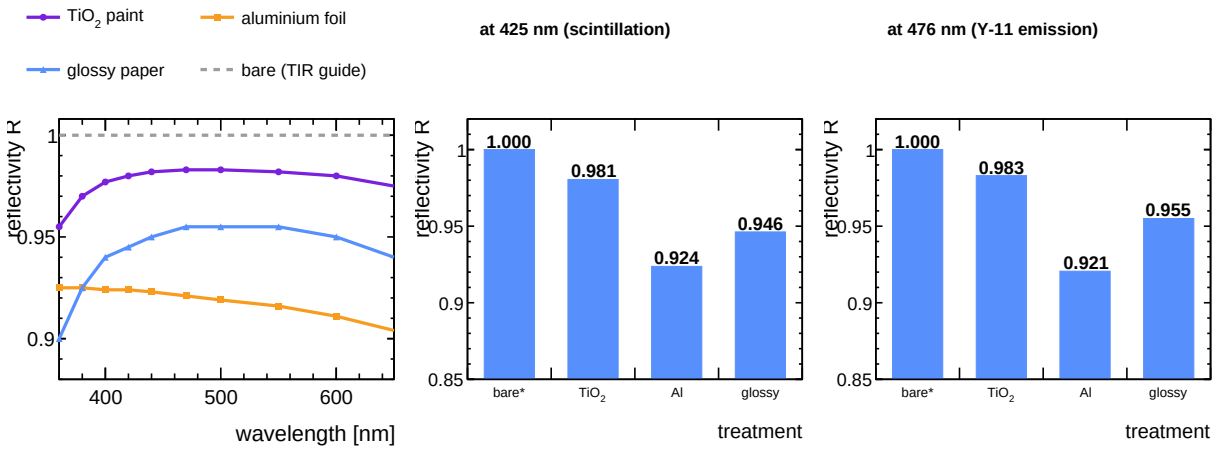


Figure 6: Reflector treatments. *Left:* tabulated reflectivity $R(\lambda)$ of the three reflective wrappers (TiO_2 paint, aluminium foil, high-gloss paper) on the shared 10-point grid, with the bare plate's lossless total-internal-reflection guide line. *Middle:* R at the scintillation emission peak (425 nm). *Right:* R at the Y-11 emission peak (476 nm), the wavelength of the config-4 re-emitted light. In both bar panels the bare plate is shown at unity as the TIR ceiling, not as a reflector.

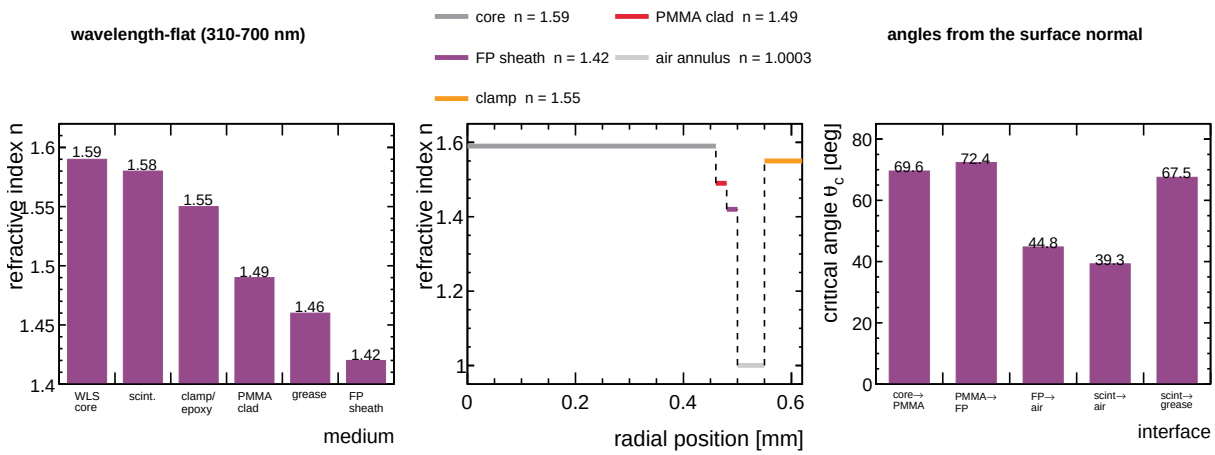


Figure 7: Refractive-index structure of the optical model. *Left:* the index ladder of the bulk media, all wavelength-flat across the visible—fibre core, scintillator, clamp/holder/epoxy, PMMA inner cladding, optical grease and fluoropolymer sheath. *Middle:* the radial index profile of the multi-clad Y-11 fibre from the core axis outward: core ($n = 1.59$), PMMA cladding ($n = 1.49$), fluoropolymer sheath ($n = 1.42$), the deliberate 0.05 mm air annulus, and the surrounding clamp ($n = 1.55$). *Right:* the critical angles $\theta_c = \sin^{-1}(n_2/n_1)$ of the successive layer pairs, measured from the interface normal. The large thresholds at the core–clad and clad–sheath boundaries are what sustain core–cladding total internal reflection, while the plate's air interface sets the usual escape cone.

2.7 Optical physics implementation

Optical photons are transported by the standard Geant4 optical physics module [4], with scintillation, Čerenkov, bulk absorption, Rayleigh scattering and boundary processes all active. Multiple scattering (Mie) is disabled; wavelength shifting (the `G4OpWLS` process) is enabled only for the config-4 fibre layout (Sec. 2.4) and inert for all others. This subsection describes the boundary physics and the photodetection model, and states explicitly the Geant4 behaviour that each modelling choice relies on, verified against the Geant4 source in `.refs/source/processes/optical`.

2.8 The boundary process and the unified surface model

Photon crossing of a dielectric interface is handled by `G4OpBoundaryProcess`, which selects an `G4OpticalSurface` from the unified surface model of Levin and Moisan [5] (whose key parameters for Geant4/GATE photon transport were later characterised [6]), using (*i*) a `G4LogicalBorderSurface` between two physical volumes or (*ii*) a `G4LogicalSkinSurface` on a logical volume. Reading the source (lines 319–347), the lookup is: a border surface between the pre- and post-step volumes is consulted first; if none exists, the skin surface of the daughter (or, failing that, the mother) logical volume is used. This establishes two facts on which the geometry relies:

1. a `G4LogicalBorderSurface` takes precedence over a `G4LogicalSkinSurface` at the same boundary—hence the grease and SiPM patches punch "holes" in the wrapping skin;
2. if no surface object exists at a boundary, the process falls back to the default lossless TIR behaviour (via `DielectricDielectric()`).

The boundary process requires a valid `RINDEX`: on medium 1 always, and on medium 2 for a `polished`/`ground` `dielectric_dielectric` boundary (lines 289–306, 436–453); `*backpainted` finishes instead read the backing index from the *surface* MPT (lines 356–374). If the property is absent the process sets status `NoRINDEX` and kills the photon *silently*—the message is printed only above verbose level 1—which is why every material (and every backpainted surface) in the model carries an explicit `RINDEX` table. This silent kill is also why the far-end fibre mirror needed an explicit skin (Sec. 2.4): the metal cap has no `RINDEX`, so the un-covered core-to-cap crossing was being lost.

For the *unified* model the probability weights for the reflection sub-models are read from the surface MPT—`SPECULARSPIKECONSTANT` (f_{ss}), `SPECULARLOBECONSTANT` (f_{sl}) and `BACKSCATTERCONSTANT` (f_{bs})—together with `REFLECTIVITY` (line 381) and, where applicable, `TRANSMITTANCE` (line 394) and `EFFICIENCY` (line 390). The surviving probability $1 - f_{ss} - f_{sl} - f_{bs}$ is shared between diffuse (Lambertian) and bare reflection.

2.9 Reflector finishes

The four treatments use three distinct code paths in `G4OpBoundaryProcess`, which we verified:

- `groundfrontpainted` (TiO_2). The finish is handled by the "painted-front" branch (lines 462–488): a uniform random draw against `REFLECTIVITY` decides absorbed / transmitted / reflected, and reflection is set to a pure Lambertian distribution (`LambertianReflection`, line 481). There is *no* TIR at this finish—photons are either diffusely reflected or absorbed. This is the physically correct description of sub-surface scattering in a thick TiO_2 paint.
- `polishedbackpainted` (Al, glossy paper). This finish is routed to `DielectricDielectric()` (line 456–459). The scintillator/backing interface still exhibits Fresnel reflection and TIR; sub-critical photons cross it and are returned by the backing "paint" with reflectivity $R(\lambda)$ and a reflection distribution set by f_{ss}, f_{sl}, f_{bs}. We choose $f_{ss} = 1$ (pure specular spike), since both wrappers are smooth mirrors, and set the backing refractive index to distinguish the two physical situations. For the aluminium foil the backing index is the coupling grease ($n = 1.46$), following Ref. [1], which applies grease so the foil optically contacts the plate; the critical angle is then only $\arcsin(1.46/1.58) = 67.5°$, so most light refracts out of the plate and is returned specularly by the foil, whose reflectivity is the tabulated normal-incidence reflectance of evaporated aluminium. For glossy paper the backing index is that of air ($n = 1.0003$), so the plate keeps its full TIR cone and only the sub-critical fraction reaches the paper, which acts as a slightly lossier mirror.
- **Bare plate**. This is deliberately *not* given a wrapping skin. The plate keeps its plain polished boundary across a real 1.2 mm air gap to the holder shell. The sub-critical fraction refracts across that gap and is then killed by the holder: the black shell carries an absorbing skin ($R = T = 0$), and its bulk (1 µm absorption length) would absorb the light anyway. Super-critical light is retained by TIR. There is *no* absorbing skin on the plate itself; the same air-gap loss is applied at the read-out pads, which are separated from the plate by a thin air border in this configuration.

2.10 Wavelength shifting

In the config-4 layout, photons that reach the fibre cores are absorbed and re-emitted isotropically at longer wavelength by the `G4OpWLS` process, which is configured through the fibre material's `WLSABSLENGTH`, `WLSCOMPONENT` and `WLSTIMECONSTANT` properties. Both spectra are the digitized Kuraray Y-11(200) curves: the absorption band peaks at 430 nm (onset $\approx$ 486 nm) and the re-emission peak is 476 nm (Fig. 4). The absolute absorption strength is set by the 430 nm absorption length, 1.0 mm in production (with a systematic band of 0.3 mm to 3 mm available through the `setWLSAbsLength` command). The re-emission is sampled with an exponential time profile, $\tau = 7$ ns, rather than a fixed delay. `G4OpWLS` samples the secondary from `WLSCOMPONENT` subject to $E'_\gamma \leq E_\gamma$, so the emission table is cut at the absorption onset; a small overlap of the emission tail with the absorption band gives mild self-absorption. The process is switched on by the physics list when the WLS geometry is detected (the Geant4 optical-physics constructor reads the process activation at build time, so the flag is set before the base construction). Because most of the shifted emission lies red-ward of the absorption band, the re-emitted light is transported along the fibre by TIR to the read-out SiPM—which is why the detected config-4 spectrum is green-shifted (mean $\sim$510 nm) relative to the direct-coupled configurations.

2.11 Photodetection: the SiPM photocathode

Photodetection is delegated entirely to Geant4, rather than being drawn manually in the sensitive detector. The grease/SiPM boundary is a `dielectric_metal` surface whose MPT carries the SiPM `EFFICIENCY` as a function of photon energy (equivalently, of the wavelength λ), i.e. the photon detection efficiency (PDE) of an ASD-NUV4S-P-40 device [10], peaking at 0.431 near 410 nm (the digitized 25-point curve of Fig. 4; values fall to ≈ 0.30 at 490 nm), with `REFLECTIVITY` $= 0$.

The mechanism is worth stating precisely because it is the reason the sensitive detector needs no random draw of its own. In `DielectricMetal()` (lines 724–817), when the Fresnel–PDE roll places a photon in the "absorbed" channel, `DoAbsorption()` is called. Inside it (header, lines 305–320):

$$f_{\text{status}} = \begin{cases} \text{Absorption}, & \xi \geq \text{EFFICIENCY}(\lambda), \\ \text{Detection}, & \xi < \text{EFFICIENCY}(\lambda), \end{cases} \qquad \xi \sim U(0,1), \tag{1}$$

where EFFICIENCY(λ) is the tabulated PDE. The absorbed photon is assigned a non-zero `LocalEnergyDeposit` *only* when the PDE draw succeeds. Back in `PostStepDoIt`, if the status is `Detection` *and* the invoke-SD flag is set, the sensitive detector is invoked (lines 559–560):

$$f_{\text{status}} = \text{Detection} \ \wedge \ f_{\text{InvokeSD}} \implies \text{InvokeSD(step)}. \tag{2}$$

The flag is enabled in the physics list via `G4OpticalParameters::SetBoundaryInvokeSD(true)`. Consequently the sensitive-detector `ProcessHits` fires *only* for genuine photoelectrons (it tests that the step energy deposit is positive), counts one photoelectron per call, and kills the photon (the SiPM is opaque). This is a clean separation of concerns: the physics of the photocathode lives in the optical surface, and the sensitive detector only books hits.

2.12 Per-event optical accounting

Two per-event counts recur throughout this report. Write $N_{\text{pe}}^{(e,s)}$ for the number of photoelectrons detected in SiPM s during event e, and $N_{\gamma}^{(e)}$ for the number of optical photons generated in the slab during event e (scintillation plus Čerenkov), irrespective of their later fate.

A thread-local diagnostic record (`OpticalDiag`) accumulates, per event: scintillation photons, Čerenkov photons, terminated photons, detected photoelectrons, and photons that escaped to the world. The primary muon's first-entry position in the slab is captured on the first step inside the scintillator logical volume and stored in slab-local coordinates. At end of event the sensitive detector writes one row per (event, SiPM), *including zero-detection rows*, so the sample is unbiased in the muon interaction position. Each row carries the entry position, the event's produced-photon count (the efficiency denominator), the detected count in that SiPM, the first arrival time, the mean and RMS detected wavelength, and the track weight. Three variable-length branches additionally list the individual photoelectrons: their global arrival times (`tDetected`), wavelengths (`wlDetected_nm`) and creating processes (`creatorDetected`, with 1 = `OpWLS`, 2 = scintillation, 3 = Čerenkov). The per-row count, timing and spectral data therefore support per-SiPM, combined and spatially-binned collection-efficiency analyses together with offline pulse-shape and process-mix studies.

Figure 8 shows a head-less event display of a single vertical muon in the glossy-wrapped plate (config 1). For legibility the display uses a 30 MeV muon—just below the Čerenkov threshold of the scintillator—with the scintillation yield rescaled, so that the individual optical tracks can be followed: photons are generated along the muon path, propagate by total internal reflection and specular reflection off the wrapper, and terminate on the SiPM patches at the four edges.

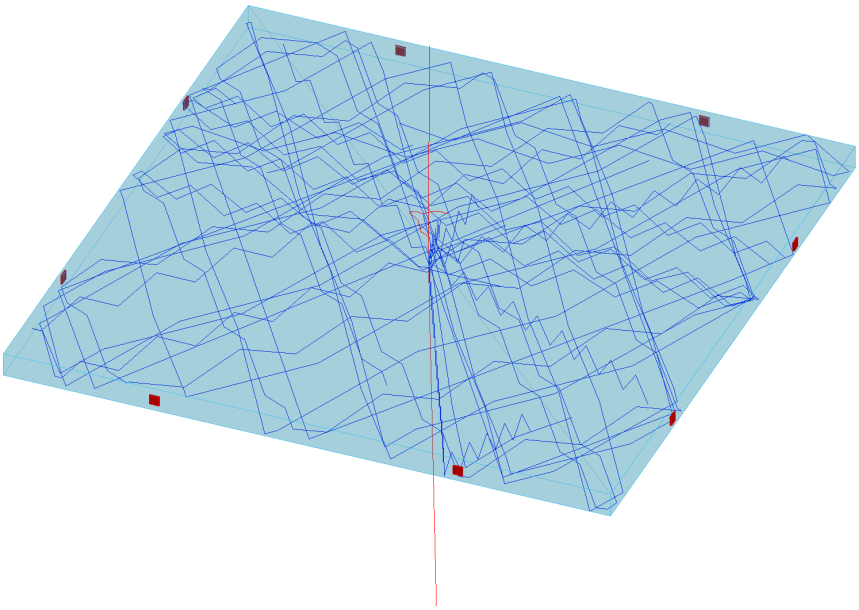


Figure 8: Head-less (`TSG_OFFSCREEN`) event display of one vertical muon in the glossy-wrapped plate with the config-1 layout, on a white background. The red line is the muon and the blue lines are optical photons (here purely scintillation, since the muon is below the Čerenkov threshold); the plate and its reflective wrapper are drawn as translucent surfaces with their edges outlined, and the SiPMs are opaque red. Both the muon energy (30 MeV, below the Čerenkov threshold) and the photon yield are chosen for visualisation only, so that individual tracks are resolved rather than merging into a saturated blob. The display is generated reproducibly by `macros/muon_vis.mac`.

3 Simulation campaigns and surface calibration

3.1 Beam, irradiation patches, and efficiency definition

The primary source is a vertical muon pencil beam: a 50 mm×50 mm square beam of 4 GeV μ^- normally incident on the plate, with the beam centre placed at three positions, each irradiating a 50 mm×50 mm patch lying *entirely on* the polished plate. With the plate spanning $[-125, 125]$ mm in both x and y, the patch centres are

$$\text{centre} = (0, 0), \qquad \text{edge} = (0, -100), \qquad \text{corner} = (100, -100)\ \text{mm},$$

so the illuminated regions are $[-25, 25]^2$, $[-25, 25] \times [-125, -75]$ and $[75, 125] \times [-125, -75]$ respectively (report frame: $+y$ is the top, $-x$ the left). This places the edge and corner regions on the bottom ($-y$) edge, opposite the single instrumented face of config 3 (top, $+y$), as in Fig. 3 of Ref. [1]. It also matches the 50 mm×50 mm active area of the muon coincidence detector (MCD) that defines the regions of interest in Ref. [1], and keeps the irradiated area—and hence the denominator of Eq. 3—equal at all three positions. Each (config, reflector, position) runs 2500 events after a single `/run/initialize`. The matrix therefore comprises

$$4\ \text{configs} \times 4\ \text{reflectors} \times 3\ \text{positions} = 48\ \text{files}, \qquad 48 \times 2500 = 120\,000\ \text{events}.$$

Two focused calibration campaigns bracket the remaining surface inputs and are presented first. A bare-contact scan runs configs 1–3 at eleven contact fractions $f = 0, 0.005, \ldots, 0.050$ at the three beam positions, $3 \times 11 \times 3 = 99$ files with 1000 events each (99 000 events); an aluminium-backing scan runs configs 1–3 with `grease` and `air` backing at the three positions, $3 \times 2 \times 3 = 18$ files with 1000 events each (18 000 events). The Al scan is taken from the earlier production campaign

because the Al model is unchanged and that scan is complete; the contact scan is the new extended campaign. Together with the whole-plate run of Sec. 5 ($16 \times 25\,000 = 400\,000$ events), the campaign totals 637 000 events, of which 237 000 are matrix-style (matrix + contact + Al). The next two subsections present the calibrations and select the nominal surface parameters; the production matrix and whole-slab results follow in Secs. 4 and 5.

Production runs were executed on the HPC cluster. The optional per-slab optical track-length flux map is disabled for the matrix (it is retained as a separate diagnostic); the collection efficiency is reconstructed entirely offline from the `sipmHits` tree at any desired binning, so the map is not needed.

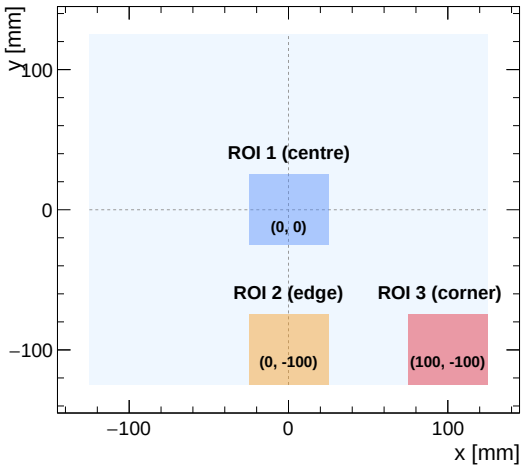


Figure 9: Plan view of the three irradiation sites (regions of interest) on the 250 mm×250 mm plate, in the report frame ($+y$ top, $-x$ left). Each coloured square is the 50 mm×50 mm footprint of the vertical muon beam: ROI 1 (centre, blue) at $(0, 0)$, ROI 2 (edge, orange) at $(0, -100)$, and ROI 3 (corner, red) at $(100, -100)$. The edge and corner sites lie on the bottom ($-y$) edge, opposite the single instrumented face of config 3.

The collection efficiency is defined, for a set $\mathcal{S}$ of SiPMs and an event e, as

$$\varepsilon = \frac{\sum_{e} \sum_{s \in \mathcal{S}} N_{\mathrm{pe}}^{(e,s)}}{\sum_{e} N_{\gamma}^{(e)}}, \tag{3}$$

where the outer sum runs over all simulated events and the inner sum over the SiPM set $\mathcal{S}$. The all-SiPM sum over the three beam positions is the per-position figure of merit. A *plate-averaged* efficiency is the appropriate global figure of merit, but the three patches sample only part of the plate; it is therefore obtained from the uniform whole-plate irradiation of Sec. 5 and reported there.

For presentation, all position-resolved results (the maps of Fig. 9 and the whole-slab maps of Sec. 5) are shown in the report frame with the top edge at $+y$, the bottom at $-y$, the left at $-x$ and the right at $+x$. The internal simulation frame is this view rotated by 90° clockwise; the primary muon entry points and the SiPM markers are mapped into the report frame by $(x, y) \to (y, -x)$ before plotting. This rotation is a rigid change of view and leaves every scalar result (yields, spectra, efficiencies) unchanged; it only fixes the orientation of config 3 (top edge) and of the edge/corner irradiation sites (bottom edge).

The optical measurements do not fix two remaining surface inputs: the effective bare-holder contact fraction f and the aluminium backing. We determine both before presenting any production result, so that every matrix and whole-slab yield below uses one consistent nominal surface model. Both calibrations use the same beam patches and efficiency definition just described.

3.2 Bare-plate holder contact

The clean bare model assumes a 1.2 mm air gap between the polished plate and the black holder. A real clamped plate touches the holder at scattered high spots, and each contact frustrates total internal reflection and removes trapped light. To bracket this unmeasured loss channel we repeat the three-patch scan for a mean-field contact fraction f: every hit on a bare face is absorbed with probability f and otherwise follows the normal Fresnel/TIR path, implemented as a polished `dielectric_dielectric` skin with `REFLECTIVITY` $= 1 - f$ and `TRANSMITTANCE` $= 0$ (Sec. 2.3.1). The extended scan covers configs 1–3, the three beam positions and eleven fractions $f = 0$, 0.005, ..., 0.050, with 1000 events per case (99 files, 99 000 events). It is an effective area fraction, not a measurement of the holder gap.

Figure 10 shows the absolute yield versus f, with the matrix TiO_2 reference (reported in Sec. 4) drawn as dashed lines, and the normalised centre yield. At $f = 0$ the config-1 bare plate collects 304.95, 295.37 and 288.55 pe/event at the centre, edge and corner; the corresponding config-2 values are 160.21, 153.68 and 152.25, and the config-3 values 81.27, 79.20 and 80.23. A 1% contact fraction already removes about half the light: at the config-1 centre the yield falls to 155.01 pe/event, and at the nominal $f = 0.045$ to 74.37. The loss is therefore strongly non-linear in the first few percent of contact area, as expected when a small fraction of the surface provides a fast escape path for the trapped mode volume.

The nominal value is chosen from the crossings themselves. At the centre the crossing is remarkably stable across layouts, $f = 0.0165$–0.0174 for configs 1–3. At the edge and corner the required fraction drops to 0.0033–0.0088 in the four-sided layouts, where the diffuse TiO_2 reference is bright, while in the single-sided config 3 it rises to 0.0389 (edge) and 0.0412 (corner), because the TiO_2 edge and corner patches there are the dimmest of the three positions. The smallest sampled fraction that places all nine cells below TiO_2 is therefore $f = 0.045$; at that value the nine bare yields are 74.37/64.42/63.25 (cfg 1), 37.60/32.35/32.04 (cfg 2) and 19.22/16.32/16.04 (cfg 3) for centre/edge/corner. The same global f thus matches the ordering in every layout and position, at the cost of over-suppressing the cfg 1/2 centre relative to the calibration target; we retain $f = 0$ as the clean-air reference and lower bound on the loss.

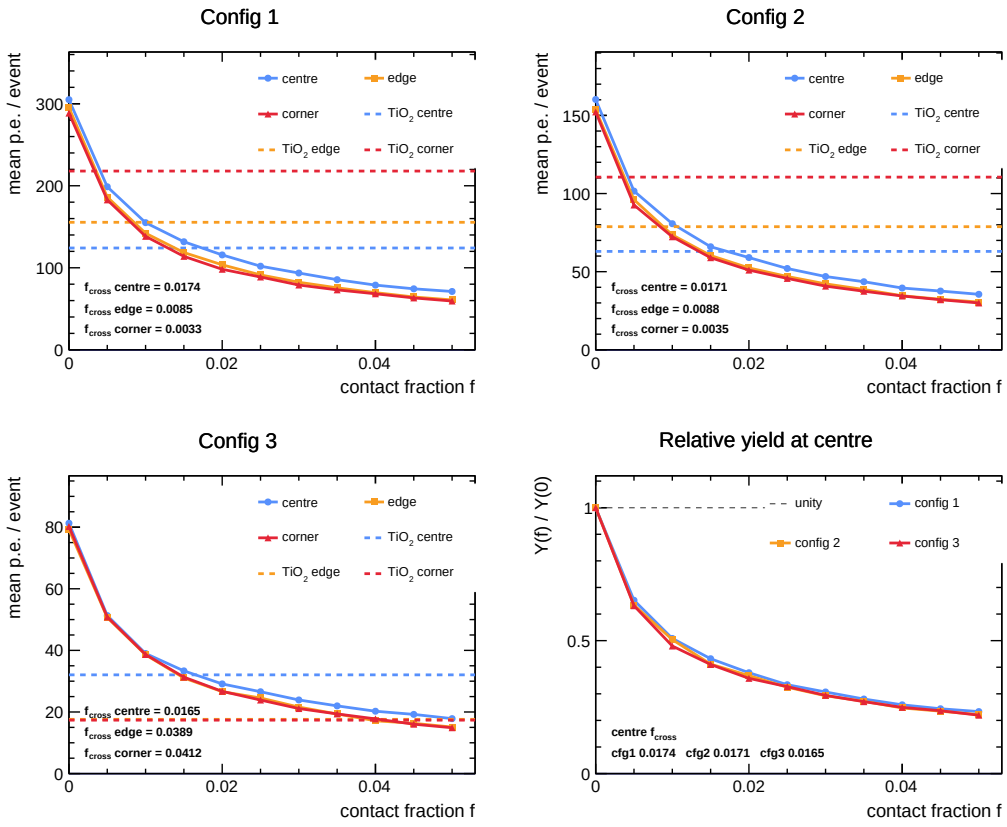


Figure 10: Bare-plate contact-fraction scan. The first three panels show the mean photoelectron yield versus contact fraction f for configs 1–3, with the centre, edge and corner patches as separate curves and the corresponding matrix TiO_2 reference as a dashed line in the same colour. The bottom-right panel shows the centre yield normalised to $f = 0$. Markers in the first three panels indicate the interpolated crossing where the bare yield falls below TiO_2. The contact model is a mean-field per-hit loss ($R = 1 - f$, $T = 0$), so f is an effective area fraction, not a measured holder contact area.

Table 3 lists the config-1 centre scan. The crossing fractions, where the bare curve falls below the matrix TiO_2 value, are collected in Table 4. At the centre the crossing is remarkably stable across layouts, $f = 0.0165$–0.0174 for configs 1–3. At the edge and corner the required fraction drops to 0.0033–0.0088 in the four-sided layouts, where the diffuse TiO_2 reference is bright, while in the single-sided config 3 it rises to 0.0389–0.0412, because the TiO_2 edge and corner patches there are the dimmest of the three positions. No single f matches every cell exactly; the production nominal $f = 0.045$ is the smallest sampled fraction that places all nine cells below TiO_2, at the cost of over-suppressing the four-sided centre. The clean-air value $f = 0$ is retained as the lower-bound reference.

Table 3: Config-1 centre contact-fraction scan. N_{pe} is the mean photoelectron yield per event with its statistical uncertainty, $Y(f)/Y(0)$ is normalised to the air-gap value, and Δ is the signed difference from the matrix TiO_2 reference (124.12 pe/event). The production nominal is $f = 0.045$.

f	N_{pe}	$Y(f)/Y(0)$	Δ (pe)
0.000	304.95 ± 2.75	1.000	+180.83
0.005	198.78 ± 1.74	0.652	+74.66
0.010	155.01 ± 1.47	0.508	+30.89
0.015	131.75 ± 1.18	0.432	+7.63
0.020	115.67 ± 1.09	0.379	−8.45
0.025	101.94 ± 0.83	0.334	−22.18
0.030	93.61 ± 0.80	0.307	−30.51
0.035	85.51 ± 0.77	0.280	−38.61
0.040	78.87 ± 0.71	0.259	−45.25
0.045	74.37 ± 0.61	0.244	−49.75
0.050	71.12 ± 0.68	0.233	−53.00

Table 4: Interpolated contact fraction f_{cross} at which the bare yield falls below the matrix TiO_2 reference, for each layout and beam position, and the bare yield at the production nominal $f = 0.045$. The TiO_2 reference is the v6 matrix value (Sec. 4).

Config	Position	Bare at $f = 0$	TiO_2	f_{cross}	Bare at 0.045
1	centre	304.95	124.12	0.0174	74.37
1	edge	295.37	155.47	0.0085	64.42
1	corner	288.55	217.92	0.0033	63.25
2	centre	160.21	62.98	0.0171	37.60
2	edge	153.68	78.79	0.0088	32.35
2	corner	152.25	110.51	0.0035	32.04
3	centre	81.27	32.06	0.0165	19.22
3	edge	79.20	17.62	0.0389	16.32
3	corner	80.23	17.35	0.0412	16.04

3.3 Aluminium backing: grease versus air

The aluminium foil is the one wrapper whose optical contact is explicitly ambiguous in the reference construction: grease is applied in places, but wrinkles and trapped air leave other regions decoupled. The production model uses the `air` backing ($n = 1.0003$, critical angle 39.3° at the scintillator), which leaves the full TIR cone intact, so only the sub-critical fraction reaches the foil. The alternative `grease` backing ($n = 1.46$, critical angle 67.5°) lets most trapped light refract into the grease and return specularly from the foil; it is retained as the lower-bound systematic. Both use the same tabulated aluminium reflectance ($R = 0.904$–0.925). The backing scan covers configs 1–3 at the three positions, with 1000 events per case, using `/dcs-monitor/det/muonScint/setAlBacking air|grease`. It is taken from the earlier complete production campaign because the Al model is unchanged; the v6 matrix air-backed centre value (479.89 pe/event) agrees with the v5 scan value (480.05) to better than 0.1 %.

Figure 11 and Table 5 show the result. Air backing raises the aluminium yield by a factor 2.38–2.59 depending on layout and position. At the config-1 centre the foil goes from 201.92 to 480.05 pe/event, at the config-2 centre from 102.51 to 254.57, and at the config-3 centre from 51.98 to 127.90. The air-backed foil reaches 92.2 %, 93.4 % and 91.6 % of the glossy yield from the production matrix (Sec. 4) in those three cases, i.e. it recovers the measured near-equality of aluminium and glossy paper. The grease-backed branch is roughly a factor 2.4 lower—about 40 % of the glossy yield—and would put the foil below the clean-air ($f = 0$) bare plate, so the backing choice is the dominant Al systematic. The enhancement is slightly larger at the edge and

corner than at the centre for config 3, but the position dependence within each backing remains weak.

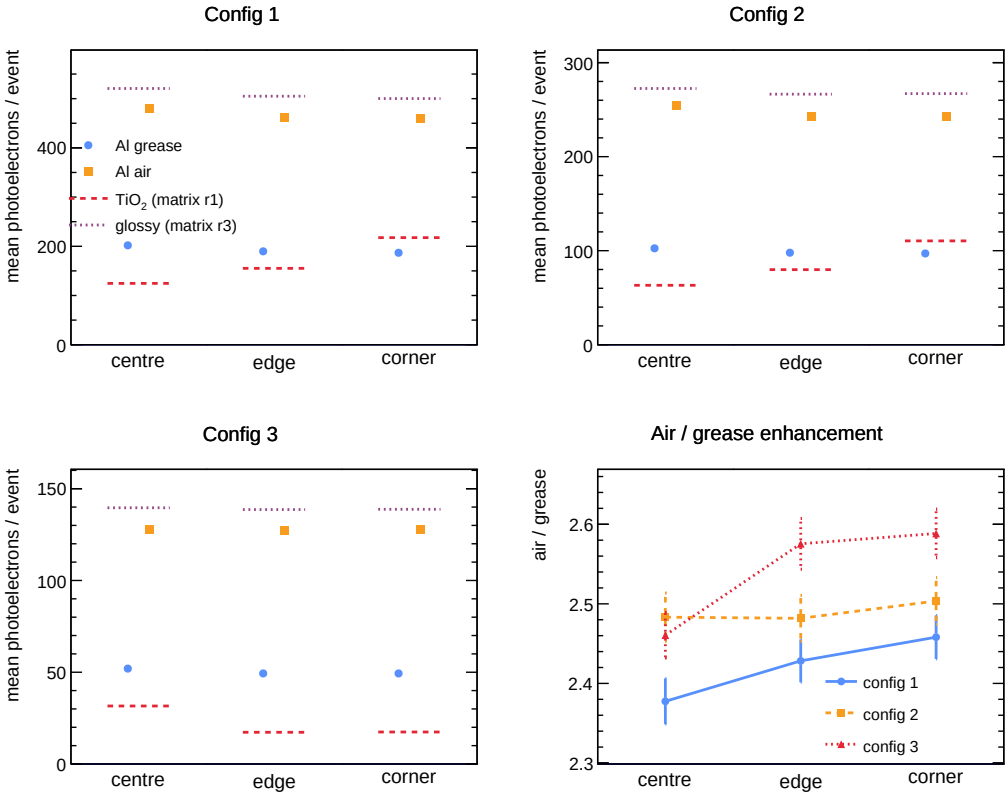


Figure 11: Aluminium backing scan. The first three panels compare the grease and air backings at the three beam positions for configs 1–3, with the matrix TiO_2 and glossy references shown for orientation; the bottom-right panel shows the air/grease enhancement ratio. The backing index changes the critical angle, not the tabulated aluminium reflectance, so the difference is entirely due to whether the plate retains its full TIR cone.

Table 5: Aluminium yield for the grease and air backings, and the air/grease ratio, for configs 1–3 and the three beam positions (1000 events per case). The production campaign uses the air backing; the grease column is the lower-bound systematic. The scan is from the earlier complete Al campaign (the Al model is unchanged); the v6 air matrix centre agrees with it to better than 0.1 %.

Config	Position	Grease (pe)	Air (pe)	Air/grease
1	centre	201.92 ± 1.79	480.05 ± 3.94	2.377 ± 0.029
1	edge	189.72 ± 1.58	460.71 ± 3.39	2.428 ± 0.027
1	corner	186.80 ± 1.45	459.18 ± 3.67	2.458 ± 0.027
2	centre	102.51 ± 0.90	254.57 ± 2.16	2.483 ± 0.030
2	edge	97.85 ± 0.86	242.85 ± 1.86	2.482 ± 0.029
2	corner	97.06 ± 0.86	243.00 ± 1.87	2.504 ± 0.029
3	centre	51.98 ± 0.45	127.90 ± 1.00	2.461 ± 0.029
3	edge	49.35 ± 0.46	127.10 ± 1.04	2.575 ± 0.032
3	corner	49.34 ± 0.44	127.71 ± 0.99	2.588 ± 0.031

The two backings bracket the as-built foil: uniform grease is the lower bound and a perfectly trapped air layer the upper bound. The air-backed result is consistent with the reference measurement's near-equality of aluminium and glossy paper. Together with the contact scan, the calibration fixes the production surface model: $f = 0.045$ for the bare plate and the air backing

for aluminium. Every matrix and whole-slab result in Secs. 4 and 5 uses these nominal values; $f = 0$ and grease are retained only as systematic bounds.

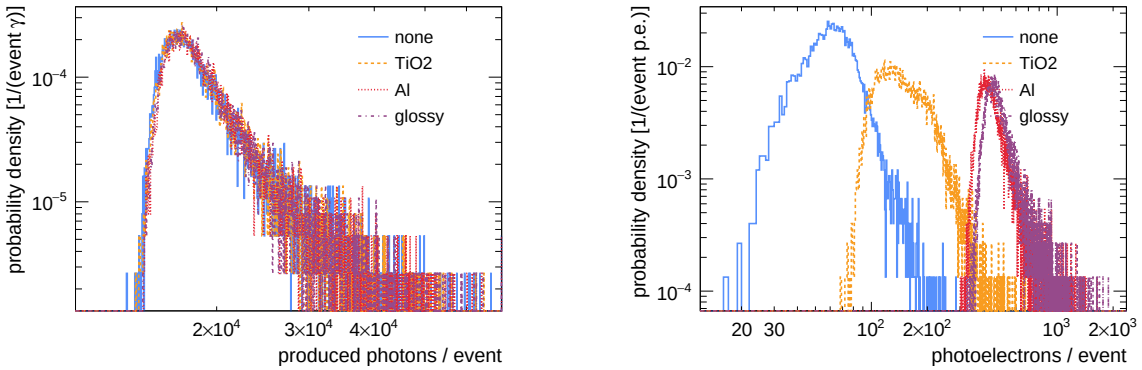
4 Matrix irradiation results

With the surface model fixed at $f = 0.045$ and air-backed aluminium, we now present the production matrix results.

4.1 Photon yield and photoelectron spectra

Figure 12a shows the per-event optical-photon yield (the denominator of Eq. 3) for the four reflectors. The mean yield is essentially independent of the reflector—about 19 600 photons per event—which is the correct physical behaviour: the reflector changes how photons are transported, not how many are produced. The small spread reflects event-to-event variation in the muon path length.

Figure 12b shows the corresponding photoelectron spectra. Here the reflector has a clear effect: at the config-1 centre the calibrated model gives 75.1 N_{pe}/event for the contact-limited bare plate, 124.1 N_{pe}/event for TiO_2, 479.9 N_{pe}/event for air-backed aluminium and 518.0 N_{pe}/event for glossy paper. The ordering is bare $<$ TiO_2 $<$ Al $<$ glossy, with the air-backed foil close to glossy paper. The bare curve is therefore a broad, low-occupancy distribution rather than the near-lossless TIR peak obtained at $f = 0$; the difference is the contact-loss systematic of Sec. 3.2.



(a) Yield (produced photons/event). **(b)** Photoelectrons/event.

Figure 12: Per-event distributions for config 1, plotted as a probability density (unit area, 1 event per unit of the abscissa) with logarithmic x- and y-axes. The production yield (left) is reflector-independent, as it must be; the detected photoelectron count (right) depends strongly on the wrapper. The log y-axis is needed because the four curves have the same event count but very different widths and heights: the narrow, low-occupancy TiO_2 distribution and the broad, high-occupancy bare and glossy distributions cannot both be resolved on a linear scale.

The efficiency scale is set by the SiPM PDE and the coupling: of order 19 600 photons are produced and the best wrapper delivers $\approx$ 519 photoelectrons, i.e. $\approx$ 2.65%. Given a PDE of $\sim$0.4 and the geometric acceptance of a side-mounted sensor, this is consistent.

4.2 Wavelength of detected photons

Figure 13 shows the wavelength distribution of detected photoelectrons, weighted by their count, for the four reflectors. The distribution peaks near 456 nm for the reflective wrappers (config-1 centre means of 456.8 nm for glossy, 456.2 nm for Al and 454.5 nm for TiO_2), i.e. slightly red-shifted with respect to the 425 nm emission peak. This is the expected consequence of self-absorption: the shorter-wavelength part of the emission overlaps the absorption band and is attenuated over

the multi-bounce path, so the surviving and detected spectrum is red-shifted. The contact-limited bare plate is bluer (449.6 nm) because the holder-contact loss suppresses the longest multi-bounce paths before they can redden the spectrum; its lower yield still leaves a well-defined distribution.

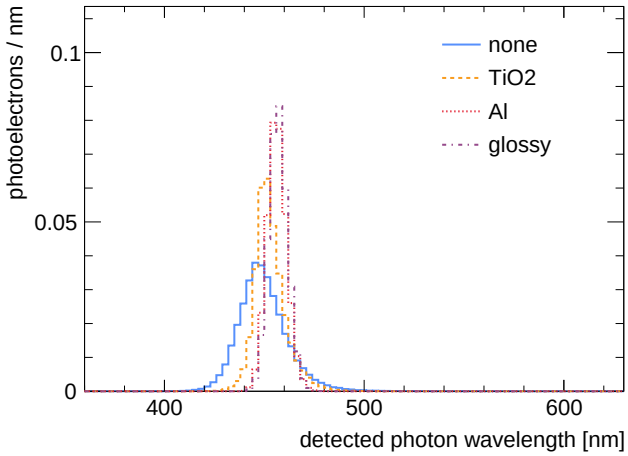


Figure 13: Wavelength distribution of detected photoelectrons (config 1), weighted by detected count. The reflective wrappers give a distribution peaking near 456 nm, red-shifted from the 425 nm emission peak by self-absorption over the multi-bounce path.

Figure 14 compares the detected-photon spectrum of the four configurations for each wrapper. Configurations 1–3, in which the SiPMs view the tile directly, all detect the same blue scintillation spectrum: the *mean* wavelength is essentially independent of the number of sensors, moving only from 457.0 nm (glossy, config 1) to 457.4 nm (config 3), the small increase reflecting the slightly longer collection path of the single-sided layouts. Config 4 is qualitatively different: the WLS fibres absorb the blue light and re-emit it at their own emission band, so the detected spectrum is shifted to 509.5 nm (glossy) — a green-shift of some 52 nm. The same shift appears for every wrapper, including the bare tile (near 509 nm), confirming that it originates in the fibre wavelength shifter rather than in the wrapper. The detected config-4 light is redder than the Y-11 emission peak (476 nm) because the blue side of the emission is preferentially re-absorbed along the fibre before reaching the read-out.

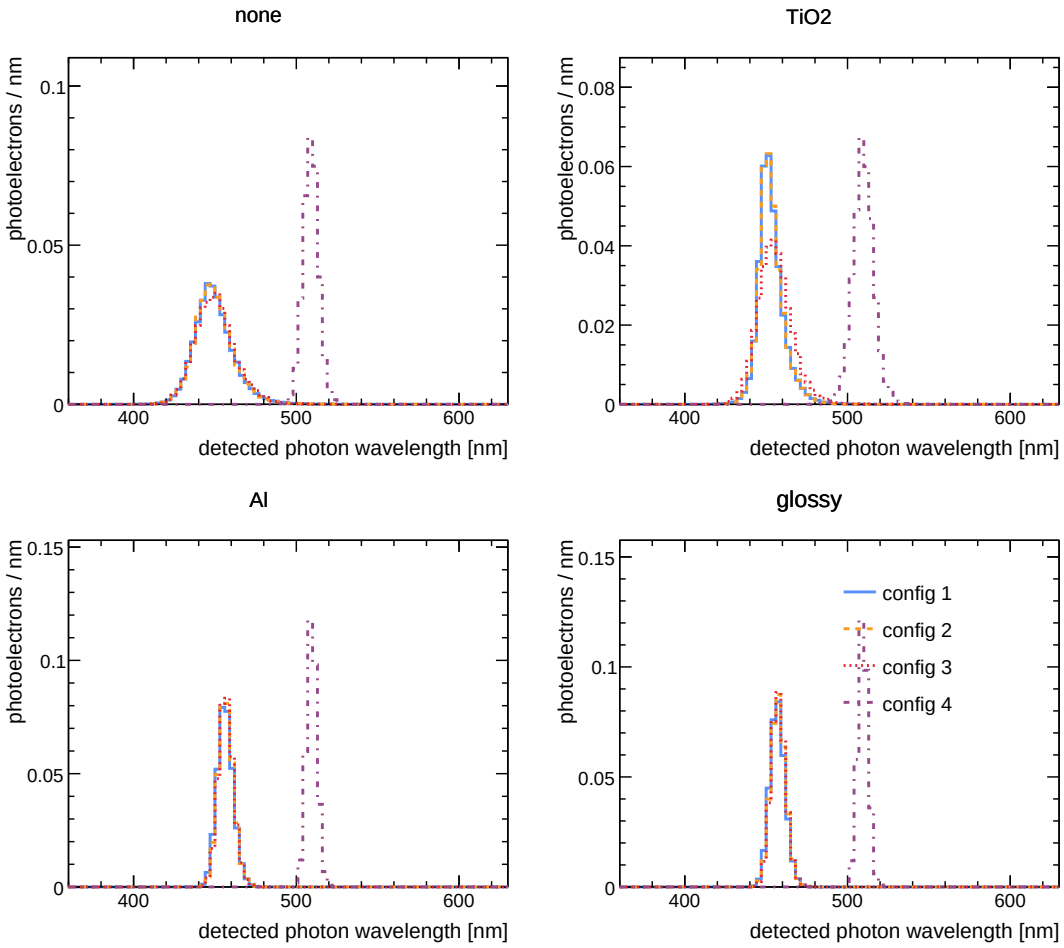


Figure 14: Detected-photon wavelength distribution, weighted by photoelectron count, for each wrapper (panel) and each configuration (curves), using the pooled data of the three irradiation positions. Configurations 1–3 follow the 457 nm blue scintillation spectrum regardless of sensor count, whereas the WLS-fibre readout of config 4 is green-shifted to ≈ 510 nm by the Y-11(200) emission band and the self-absorption along the fibre.

4.3 Spatial and directional dependence

The matrix scan separates the three 50 mm×50 mm patches (centre, edge, corner) precisely so that the response can be resolved by position. The tempting single-coordinate summary is the distance r of the patch from the plate centre, but r is blind to azimuth, and the shortcoming is decisive here: configs 3 and 4 instrument a *single* face, so the collection depends on the *direction* from which the plate is irradiated, not merely on the distance. All three patches also lie on the half of the plate opposite the instrumented face of the single-sided layouts, so their distances are not even monotonic in the proximity to a sensor. We therefore keep the three patches separate and compare them directly.

Figure 15 groups the three patches by layout (one panel per config, four reflector curves) and Fig. 16 groups them by reflector (one panel per wrapper, four layout curves). Because every patch lies fully on the plate and all three have equal area, the produced-photon denominators are equal and the three points within each curve are directly comparable. The curves are plotted on a logarithmic y-axis, which resolves the weaker TiO_2 wrapper alongside the others.

The conspicuous feature is that the specular wrappers are nearly *flat* across the three patches: across all four layouts the maximum/minimum patch ratio spans 1.5 % to 3.4 % for aluminium and 0.8 % to 3.2 % for glossy paper. The diffuse TiO_2 paint is the opposite extreme, varying by 74.8 % to 87.4 % (a factor 1.75–1.87). The contact-limited bare plate lies between them: 4.7 %

to 18.9 %, with the largest spread in the four-sided layouts. This is the expected behaviour of a closed cavity: after the first few bounces the light no longer remembers where the muon crossed the plate, so the sensor coverage, and not the irradiation point, sets the collection. The specular foil and glossy paper homogenise the response directly. The bare plate homogenises it more slowly, and the mean-field holder contact adds a position-dependent loss that is visible as the residual few-percent to 19 % spread.

These three patches lie near the *instrumented* half of the plate, so the comparison above captures the position dependence within the well-lit region. The response over the *whole* plate is resolved in Sec. 5.3; there Al and glossy are confirmed to be almost uniform, the contact-limited bare plate is intermediate, and TiO_2 retains the strongest (and now centre-depressed) position dependence.

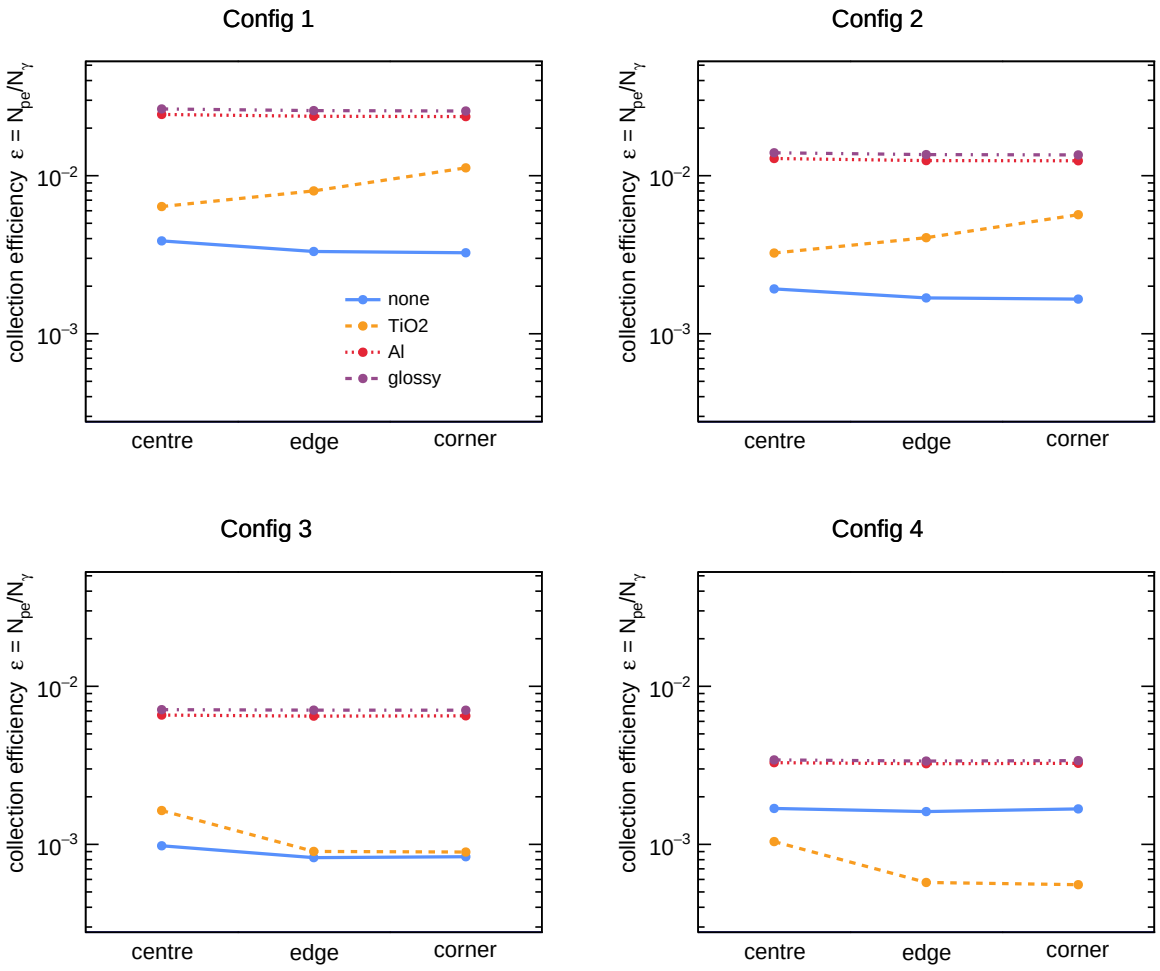


Figure 15: Collection efficiency at the three irradiation patches (centre, edge, corner), one panel per detector layout, with the four reflector curves (logarithmic y-axis). All three patches have equal area and lie fully on the plate, so the points are directly comparable. Aluminium and glossy paper are nearly flat across the three patches in every layout; the contact-limited bare plate varies by up to 19 %, and the diffuse TiO_2 paint shows by far the strongest position dependence.

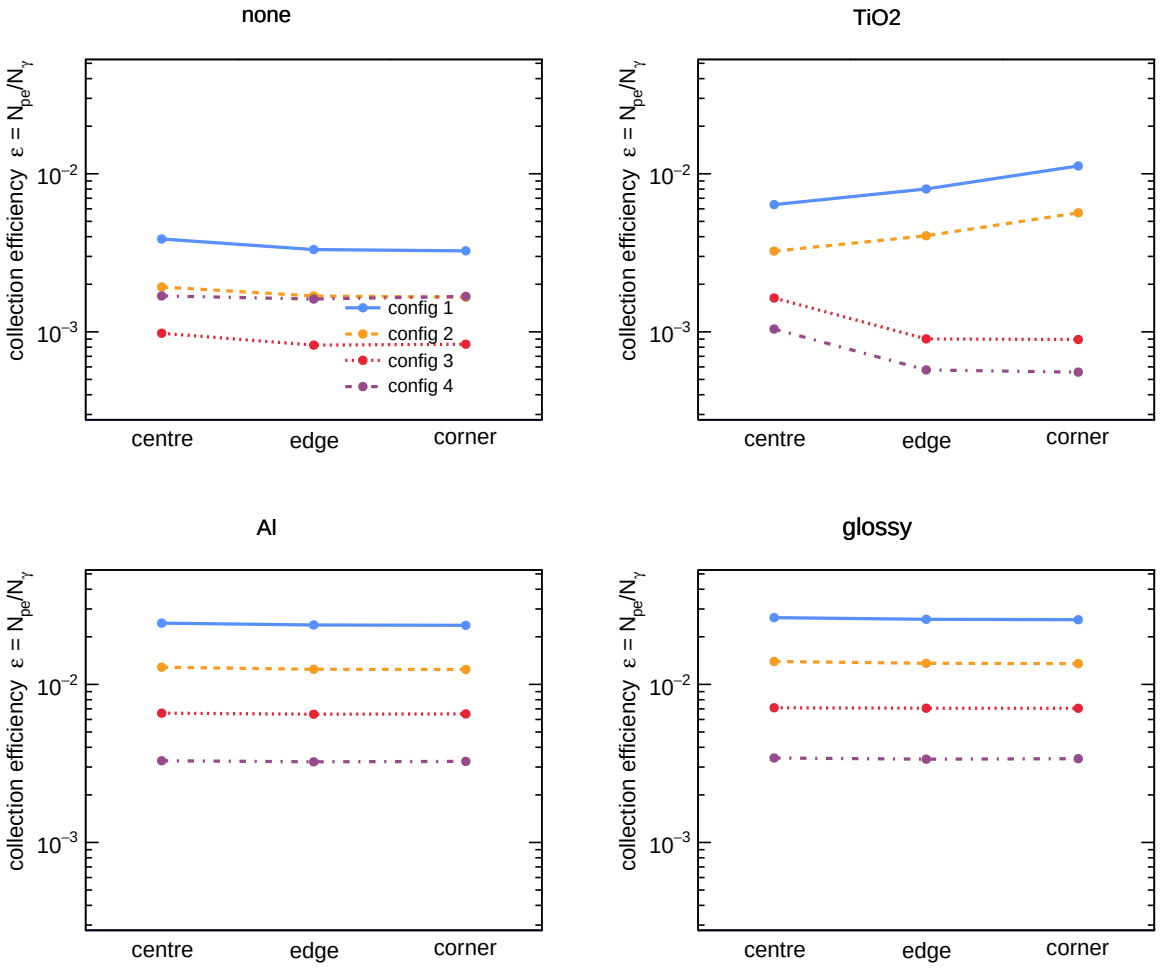


Figure 16: The same data grouped by reflector (one panel per wrapper), with the four layouts as curves. Aluminium and glossy paper are nearly flat in every layout, while the contact-limited bare plate varies by up to 19 %. The diffuse TiO_2 paint varies by a factor $\approx$1.75 in the four-sided layouts (corner highest); in the single-sided layouts (3 and 4) it drops from centre toward the edge and corner patches, which sit on the far side of the plate from its sensors.

Figure 17 resolves the directional response most directly. For config 1 (the layout with the most sensors) it shows the *share* of the collected light seen by each of the eight sensors, in one panel per reflector, with the three irradiation patches as grouped bars (dashed line: the uniform expectation of 1/8). The difference between the wrappers is instructive. Glossy paper and aluminium spread the light almost uniformly: at the corner patch the largest and smallest single-sensor shares differ by only 1.26 and 1.28. The diffuse TiO_2 paint concentrates it strongly: the largest and smallest shares differ by a factor 11.95, and the two sensors on the faces bordering the illuminated corner together collect 75 % of the photoelectrons (about three times the uniform share each), while the remaining six share the rest. The contact-limited bare plate is intermediate (8.61), a direct consequence of the position-dependent holder-loss model. The total efficiency still varies only modestly across the three patches for Al and glossy (Table 6), so the cavity keeps the *amount* of collected light nearly constant while rearranging its spatial division. A radius-only summary averages these distinct per-sensor patterns into a single number and hides precisely this information.

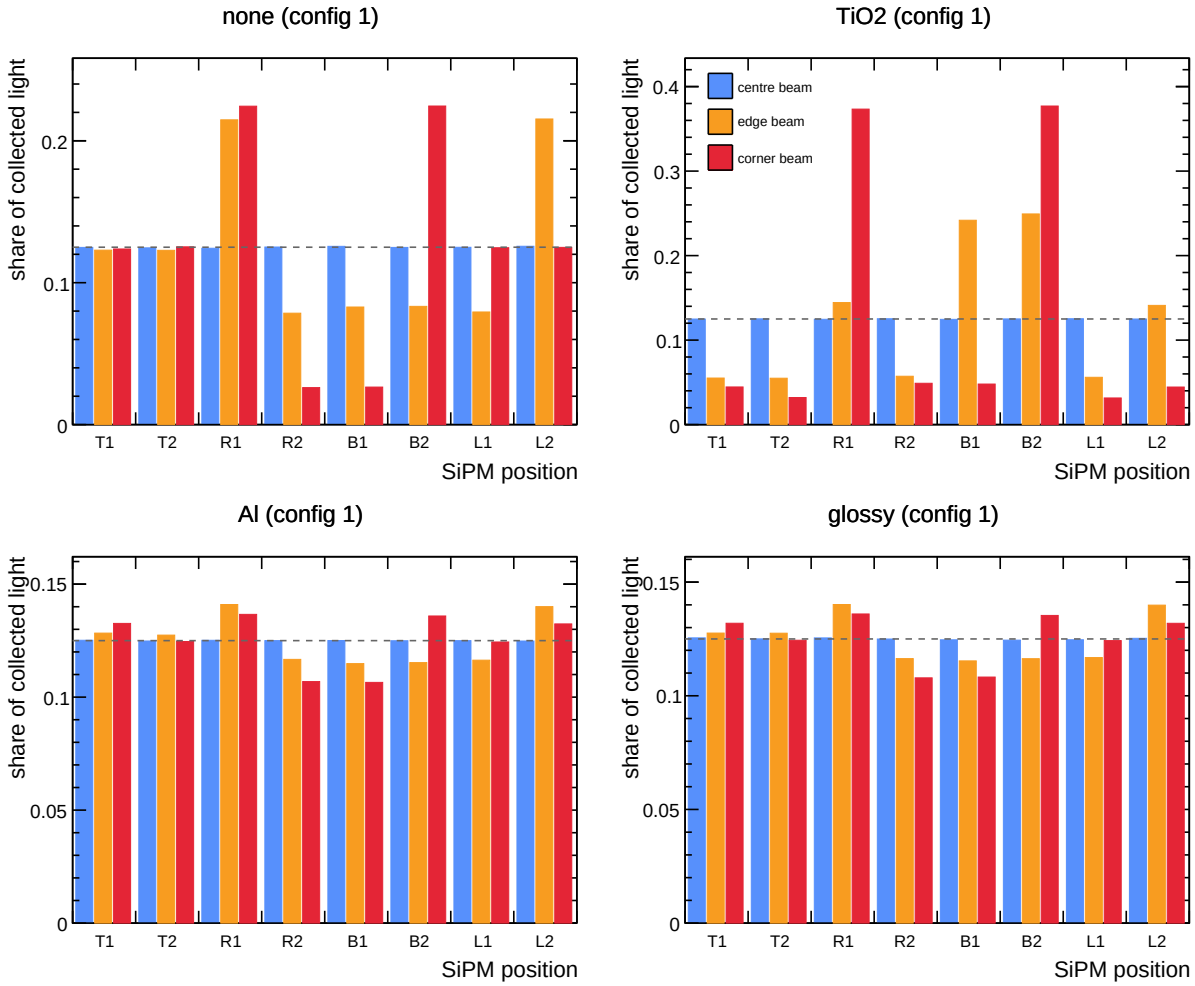


Figure 17: Direction-aware light sharing, config 1, one panel per reflector. Each group of bars is one sensor (x axis) and the three colours are the irradiation patches (blue: centre, orange: edge, red: corner); the height is that sensor's share of the light collected at that patch, and the dashed line marks the uniform expectation 1/8. Glossy paper and aluminium spread the light almost uniformly over all eight sensors (corner max/min 1.26 and 1.28), the contact-limited bare plate is intermediate (8.61), whereas the diffuse TiO_2 paint concentrates it on the two sensors nearest the beam (corner max/min 11.95).

Table 6 lists the absolute centre/edge/corner efficiencies for config 1; the fully position- and direction-resolved maps over the *entire* plate are given in Sec. 5.

Table 6: Position dependence of the collection efficiency, config 1 (all SiPMs). Each beam patch is 50 mm×50 mm and lies fully on the plate, so the produced-photon denominators are equal.

Reflector	Centre	Edge	Corner
Bare	3.868×10^{-3}	3.315×10^{-3}	3.254×10^{-3}
TiO_2	6.380×10^{-3}	8.014×10^{-3}	1.121×10^{-2}
Al	2.440×10^{-2}	2.376×10^{-2}	2.365×10^{-2}
Glossy	2.644×10^{-2}	2.581×10^{-2}	2.566×10^{-2}

4.4 Photoelectron spectra across configurations and positions

Figure 18 collects the per-event photoelectron spectra over the whole matrix scan in the layout of Fig. 10 of Rao et al. [1]: the three irradiation regions (centre, edge, corner) are the rows and the detector configurations are the columns, with the four reflector treatments overlaid in every cell. Two adaptations are made. First, the voltage pulse-height spectra of the measurement are

replaced by the simulation's number of photoelectrons per event, the quantity that is directly comparable between our model and the experiment. Second, the fourth configuration (single SiPM + WLS fibre clamp) is added as a fourth column, so that all four layouts appear in one figure on a common logarithmic axis (panel (b)) rather than in a separate figure.

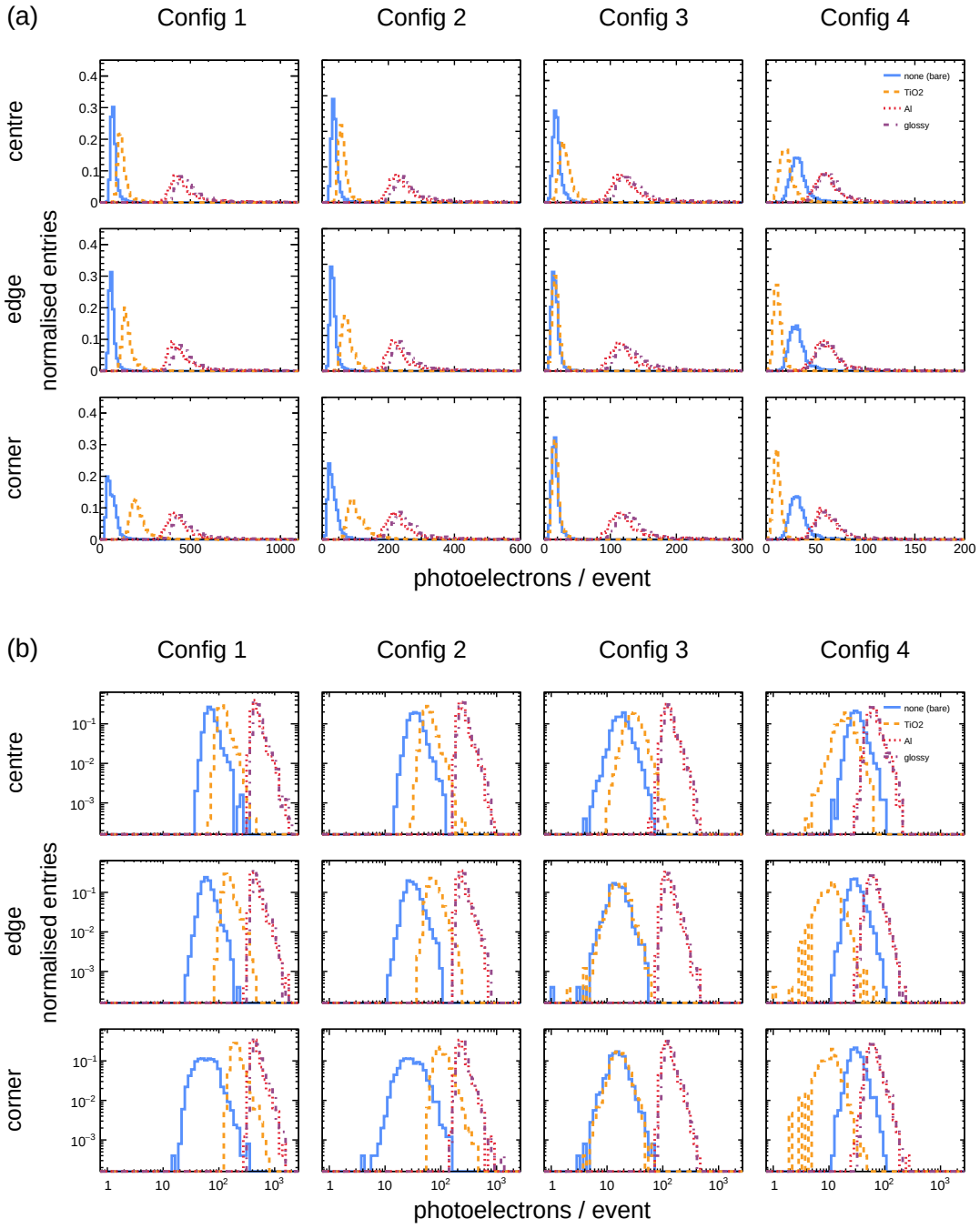


Figure 18: Photoelectron spectra in the layout of Fig. 10 of Rao et al. [1]: rows are the irradiation regions (centre, edge, corner) and columns the detector configurations (1–4, with the WLS-clamp layout as the fourth column). Each cell overlays the four reflector treatments. The histograms are normalised entry distributions (each cell is scaled so its entries sum to unity). The *upper* grid (a) uses linear axes and the *lower* grid (b) logarithmic axes; both share the same reflector colours and styles and the same units. Panel (a) gives each column its own linear abscissa, truncated at that configuration's 99.5th percentile (rounded up) and binned linearly, so that the narrow high-density core of every layout is resolved. A single shared linear abscissa would instead collapse the WLS-clamp layout (config 4), which collects roughly an order of magnitude fewer photoelectrons than the direct-coupled layouts, into the first few percent of the axis. Panel (b) shows every cell over its full distribution on a single shared logarithmic abscissa with logarithmic bins, so the four layouts remain directly comparable there. Zero-photoelectron events form a pulse below threshold and lie off the abscissa. The wrappers shift and broaden the spectrum from the low, contact-limited bare distribution through TiO_2 and air-backed Al to the wide, bright glossy one; apart from TiO_2 and the bare plate the spectra vary by at most 3.4 % across positions.

The figure condenses the trends established above. At the config-1 centre the mean yields are 75.1, 124.1, 479.9 and 518.0 pe/event for bare, TiO_2, Al and glossy, so the calibrated ordering bare $<$ TiO_2 $<$ Al $<$ glossy holds in every direct-coupled cell, with the contact-limited bare plate the lowest and glossy the highest. The position dependence is wrapper-specific: for the diffuse TiO_2 paint the corner spectrum is shifted highest in the four-sided layouts (a factor $\approx$1.75 in the mean, configurations 1 and 2) but the centre is highest in the single-sided layouts 3 and 4. Aluminium and glossy paper vary by at most 3.4 % across positions, whereas the contact-limited bare plate varies by up to 18.9 %. This is the same reflector-versus-position interplay summarised in Table 6 and the efficiency maps, presented here as the pulse-height distributions that a threshold-based read-out would actually see.

5 Whole-slab irradiation

5.1 Plate-averaged collection efficiency

The matrix scan pools only three 50 mm×50 mm patches, all on the $+x$ half of the plate, so the efficiency it yields depends on the choice of patches and is *not* a plate average. The whole-slab run removes that bias: the GPS plane is expanded to 250×250 mm (`halfx` = `halfy` = 125 mm) and 25 000 events are generated per (config, reflector), so the primary muon's first-entry position is sampled *uniformly* over the entire plate and the pooled ratio $\sum N_{\mathrm{pe}} / \sum N_{\gamma}$ is the *plate-averaged* collection efficiency—the appropriate global figure of merit. Figure 19 shows it for every (config, reflector) pair, and Table 7 lists the numbers.

With the calibrated surfaces the direct-coupled layouts reproduce the measured ordering,

$$\text{bare} < \text{TiO}_2 < \text{Al} < \text{glossy} \quad \text{(configs 1–3)},$$

while the WLS-fibre layout retains

$$\text{TiO}_2 < \text{bare} < \text{Al} < \text{glossy} \quad \text{(config 4)}.$$

Glossy paper is the brightest wrapper in every layout and air-backed aluminium reaches 92 % of its yield in configs 1–3 (96 % in config 4). The contact-limited bare plate is now the weakest direct-coupled wrapper, as the experiment finds, but it is *not* the weakest in the WLS layout: config 4 has no black holder, so the bare plate keeps its clean air gap and remains above TiO_2. The spread from weakest to brightest is a factor 6.5–7.0 in configs 1–3, set mainly by the bare contact loss, and only 2.2 in config 4. The efficiency scales roughly with the photosensitive area: the eight-SiPM layout (config 1, $\varepsilon = 4.09 \times 10^{-3}$) is highest, four SiPMs on four faces (config 2, 2.06×10^{-3}) roughly halve it, and concentrating the two sensors of config 3 on a single face (1.02×10^{-3}) halves it again. The single-SiPM WLS-fibre layout (config 4, 1.71×10^{-3}) is comparable to config 3: it is lower for TiO_2, Al and glossy by about a factor of two, but higher for the bare plate because the holder loss is absent.

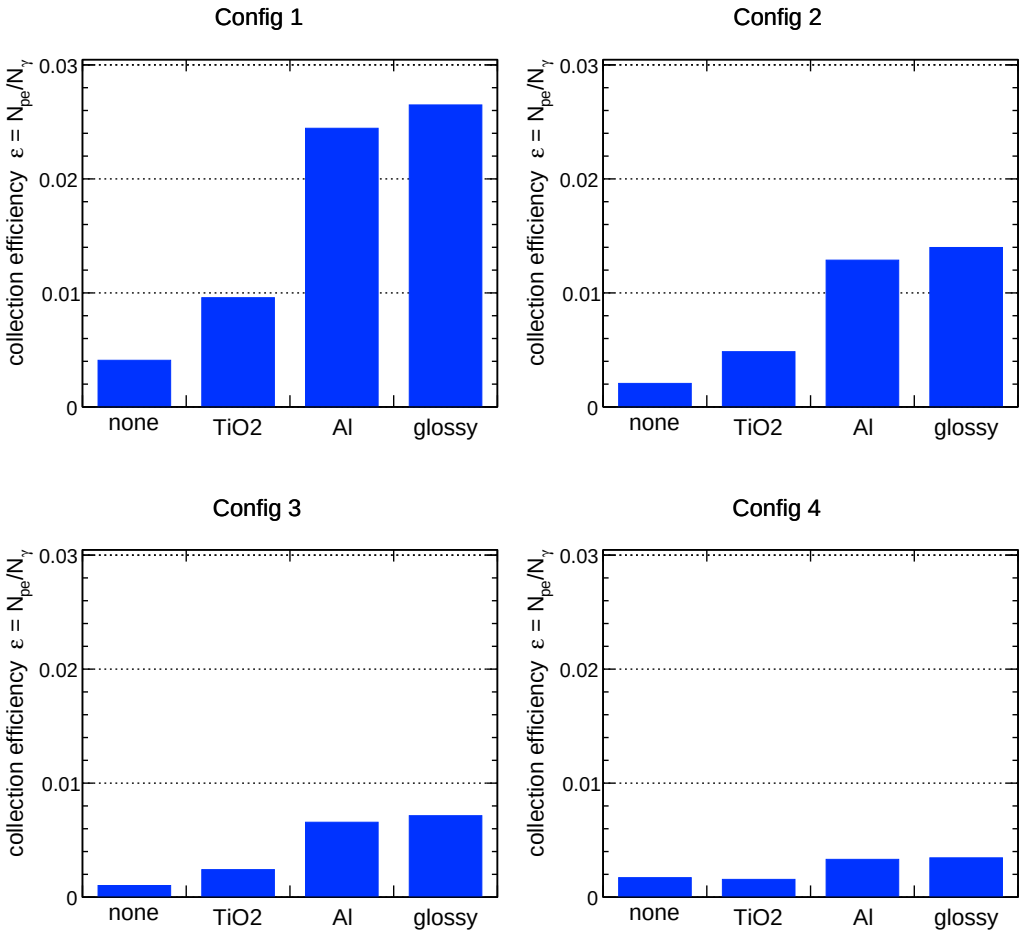


Figure 19: Plate-averaged collection efficiency $\varepsilon = \sum N_{\mathrm{pe}} / \sum N_{\gamma}$ for the four reflector treatments and four detector layouts, from the uniform whole-plate irradiation (25 000 events per panel). Glossy paper is brightest in every layout and air-backed aluminium is second in configs 1–3; the contact-limited bare plate is the weakest direct wrapper, while in the WLS layout the diffuse TiO_2 paint is the weakest.

Table 7: Plate-averaged collection efficiency (all SiPMs, uniform whole-plate irradiation; the pooled ratio of Eq. 3 over all events). The statistical uncertainties are below the quoted digits.

Layout	Bare	TiO_2	Al	Glossy
Config 1 (8 SiPMs)	4.093×10^{-3}	9.583×10^{-3}	2.444×10^{-2}	2.649×10^{-2}
Config 2 (4 SiPMs)	2.064×10^{-3}	4.847×10^{-3}	1.288×10^{-2}	1.399×10^{-2}
Config 3 (2 SiPMs)	1.022×10^{-3}	2.415×10^{-3}	6.573×10^{-3}	7.146×10^{-3}
Config 4 (1 SiPM, WLS)	1.712×10^{-3}	1.560×10^{-3}	3.317×10^{-3}	3.450×10^{-3}

A complementary, absolute figure of merit is the mean photoelectron yield per event. Because the whole-plate beam covers the entire scintillator, almost every primary enters the slab, and 399 981 of the 400 000 events register at least one photoelectron (detection probability 0.9999525). The 19 events without a detected photoelectron are 14 in config 3 bare and two in config 4 TiO_2 (zero-detection rows), plus one each in config 1 bare, config 1 TiO_2 and config 4 glossy (no `sipmHits` row, so no registered muon entry). The mean per event and the mean per *detected* event therefore differ negligibly. Table 8 lists the mean per event for every (config, wrapper) pair. Since the optical-photon production is set by the muon energy deposit and is independent of the wrapping, the yield is simply the plate-averaged efficiency of Table 7 scaled by the (layout-independent) mean number of photons produced per event, so it reproduces exactly the same

wrapper ordering.

Table 8: Mean photoelectron yield per event for the full-slab irradiation (whole-plate beam, 25 000 events per cell), for each layout and reflector treatment. Of the 400 000 whole-plate events, 399 981 detect at least one photoelectron (detection probability 0.9999525; the 19 events without a detected photoelectron are 14 in config 3 bare and two in config 4 TiO_2, plus one each in config 1 bare, config 1 TiO_2 and config 4 glossy), so the mean per detected event differs negligibly from the mean per event; the statistical uncertainties are below the quoted digits.

Layout	Bare	TiO_2	Al	Glossy
Config 1 (8 SiPMs)	79.88	186.75	477.82	518.79
Config 2 (4 SiPMs)	40.38	94.12	251.67	273.96
Config 3 (2 SiPMs)	19.97	46.95	129.11	140.21
Config 4 (1 SiPM, WLS)	33.27	30.35	64.80	67.45

5.2 Full-plate maps

The same whole-slab events that give the plate average above also resolve the response by entry position. Reconstructing the efficiency from the `sipmHits` tree in 5×5 mm cells gives the field

$$\varepsilon(x, y) = \frac{\sum_{e\,:\,(x_e,y_e)\in\text{cell}} \sum_{s\in\mathcal{S}} N_{\text{pe}}^{(e,s)}}{\sum_{e\,:\,(x_e,y_e)\in\text{cell}} N_{\gamma}^{(e)}},$$

where the sums run over the whole-slab events whose primary-muon first-entry position (x_e, y_e) falls inside the cell, i.e. identically to Eq. 3 but binned in the entry position. The whole-slab maps therefore fill in the gaps left by the three-patch scan.

Figure 20 shows the complete set of full-plate maps for config 1, one panel per reflector. The plate-averaged efficiencies are 4.093×10^{-3} (bare), 9.583×10^{-3} (TiO_2), 2.444×10^{-2} (Al) and 2.649×10^{-2} (glossy). The contact-limited bare plate has a modest inner deficit—3.839×10^{-3} within $r < 60$ mm versus 4.148×10^{-3} outside, about 7.5 %—reflecting the holder-loss model. Aluminium and glossy paper are essentially flat (2.431×10^{-2} versus 2.447×10^{-2} and 2.637×10^{-2} versus 2.652×10^{-2}). TiO_2 is the only wrapper with a strong gradient, and it runs the other way: the centre (6.738×10^{-3}) is *dimmer* than the edge and corner (1.019×10^{-2}), a factor of ~ 1.51. Diffuse recycling couples light into the sensors most efficiently when the source sits near a painted edge, and least when it is in the middle of the plate, whereas the specular transport erases the entry position almost completely.

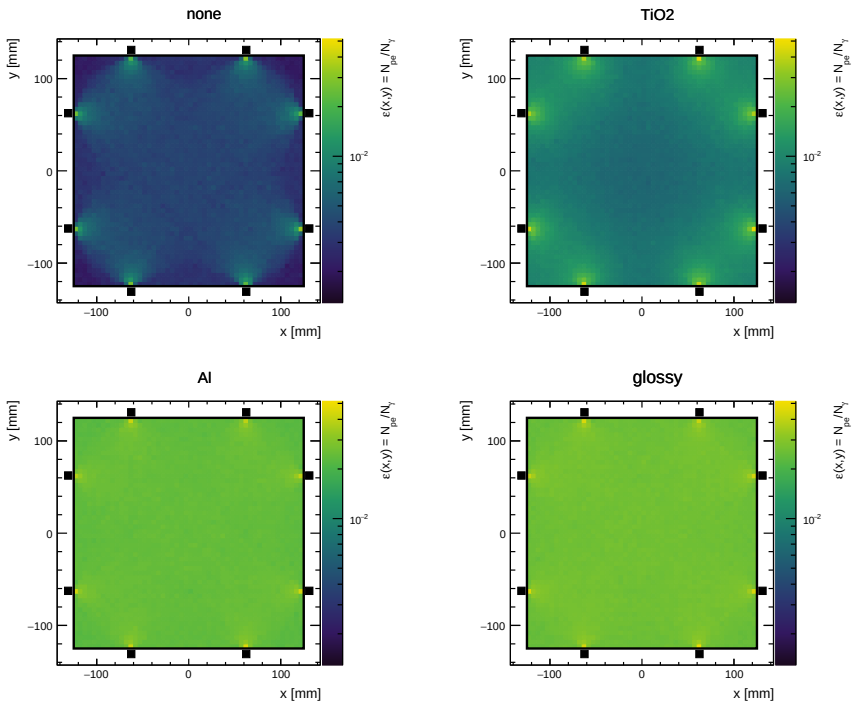


Figure 20: Full-plate collection-efficiency maps for config 1, from 25 000 whole-slab events per panel (5×5 mm cells, on a logarithmic colour scale shared by the panels). Aluminium and glossy paper are almost flat; the contact-limited bare plate (top left) is the weakest and has a modest inner deficit, while TiO_2 (top right) is the only one with a strong gradient, dimmest at the centre and brightest toward the edges and corners. The bold rectangle is the 250 mm×250 mm slab outline and the black squares mark the eight SiPM positions.

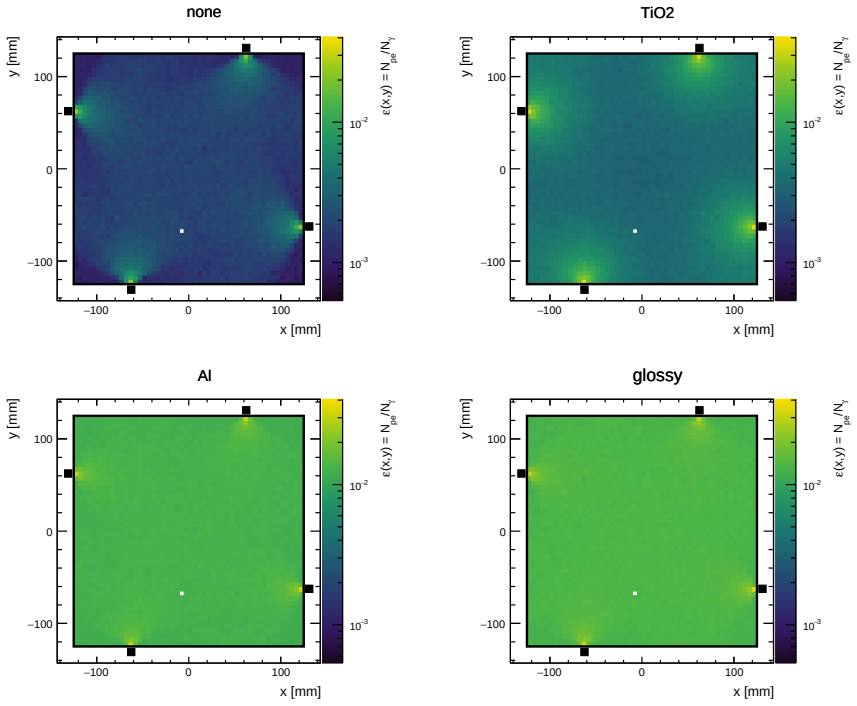


Figure 21: Full-plate collection-efficiency maps for config 2 (four SiPMs, one per lateral face at the config-1 offsets), otherwise identical to Fig. 20. The reflector ordering is the same, Al and glossy are almost flat, and each efficiency is about half the config-1 value.

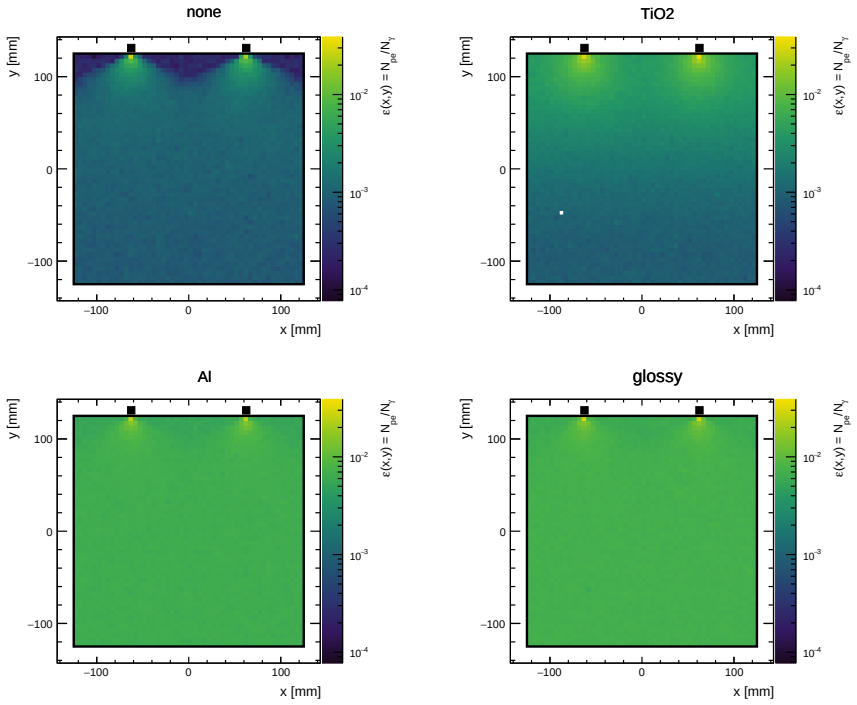


Figure 22: Full-plate collection-efficiency maps for config 3 (two SiPMs, both on the top $+y$ face). The TiO_2 panel shows the clearest top–bottom asymmetry, the contact-limited bare plate shows a weaker one, and the Al and glossy panels remain nearly flat.

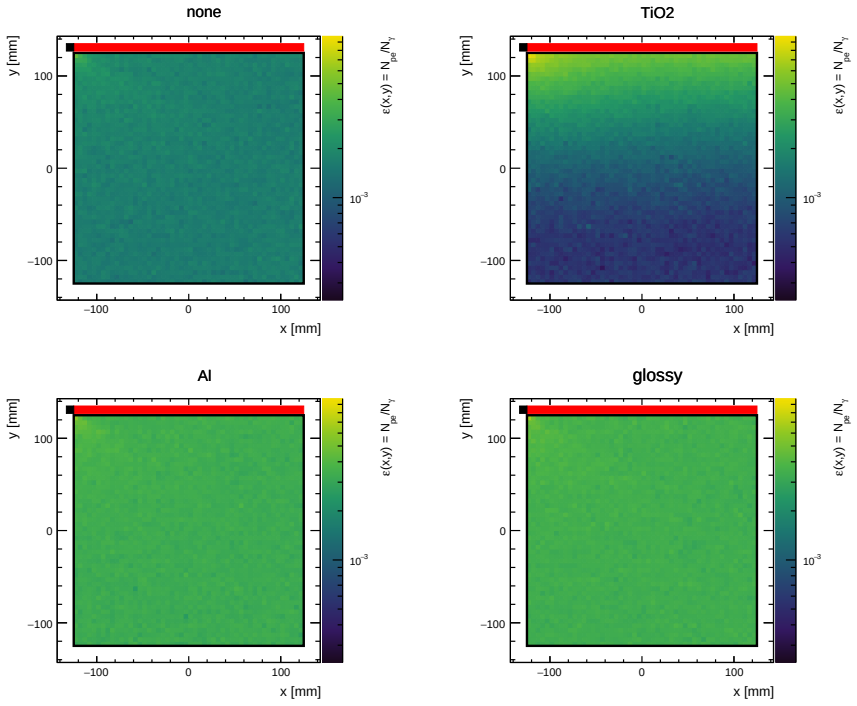


Figure 23: Full-plate collection-efficiency maps for config 4 (single SiPM + WLS fibre clamp). The efficiencies are comparable to config 3, higher for the bare plate because the black holder is absent and lower for the reflective wrappers. The TiO_2 panel retains a strong top–bottom gradient, the bare plate a weaker one, while the Al and glossy panels are almost flat. The red bar along the top ($+y$) edge marks the WLS fibre clamp (the fibres run the full plate length along it) and the black square at its left end is the single read-out SiPM, so the fibres can be located relative to the read-out. Same report frame and binning as Figs. 20–22, each panel on its own logarithmic colour scale.

Figure 21 repeats the whole-slab scan for config 2 (four sensors, one on each lateral face at the config-1 offsets). Every configuration is weaker than its config-1 counterpart—roughly halved, as expected from the reduced sensor area—while the ordering is unchanged. The bare plate gives $\varepsilon = 2.064 \times 10^{-3}$ (1.951×10^{-3} inside $r = 60\,\mathrm{mm}$ versus 2.089×10^{-3} outside); TiO_2 reaches 4.847×10^{-3} with the same centre deficit (3.409×10^{-3} versus 5.166×10^{-3}), and Al and glossy reach 1.288×10^{-2} and 1.399×10^{-2}, both nearly flat. The bare $<$ TiO_2 $<$ Al $<$ glossy sequence is therefore robust against the sensor count: fewer sensors lower the overall light collection but do not change the physics of the wrapper.

Finally, Fig. 22 shows config 3, in which both sensors sit on the same (top, $+y$) face. The integrated efficiencies are bare 1.022×10^{-3}, TiO_2 2.415×10^{-3}, Al 6.573×10^{-3} and glossy 7.146×10^{-3}, roughly half their config-2 values because concentrating both sensors on one face halves the solid angle available to photons born on the opposite side. Aluminium and glossy paper remain nearly flat (better than 1 % between the inner and outer regions). The TiO_2 panel is brightest near the instrumented top ($+y$) edge and dimmest on the far ($-y$) side: in the report frame the top/bottom efficiencies are 3.752×10^{-3} and 1.073×10^{-3}, a factor 3.50. The contact-limited bare plate shows the same sign but a much weaker asymmetry (1.165×10^{-3} versus 0.882×10^{-3}).

Figure 23 completes the set with the single-SiPM WLS-fibre layout (config 4). The efficiencies are bare 1.712×10^{-3}, TiO_2 1.560×10^{-3}, Al 3.317×10^{-3} and glossy 3.450×10^{-3}, comparable to config 3 but with the ordering TiO_2 $<$ bare $<$ Al $<$ glossy. Glossy paper is again the most efficient wrapper and air-backed Al is second. TiO_2 shows the clearest gradient: in the report frame the top/bottom efficiencies are 2.473×10^{-3} and 0.676×10^{-3}, a factor 3.66. The bare plate is much flatter (1.782×10^{-3} versus 1.642×10^{-3}), and Al and glossy agree to within about 4 %, so even with a single readout face the specular transport largely erases the entry-point dependence.

Across all four layouts glossy paper is the brightest wrapper and air-backed aluminium the second, while the weakest is the contact-limited bare plate in the three direct-coupled layouts and TiO_2 in the WLS layout. The efficiency scales roughly with the collected photosensitive area, confirming that the readout layout sets the overall scale while the wrapper sets the ordering.

5.3 Whole-slab uniformity

The full-plate maps also expose an inhomogeneity that the three discrete patches of Sec. 3 can only sample. Because the whole-slab beam irradiates every point of the plate, this inhomogeneity can be quantified *continuously* and with far better statistics.

A purely radial contrast is deliberately *not* used. The read-out layouts have no continuous rotational symmetry—configs 1 and 2 are four-fold symmetric, configs 3 and 4 single-sided—and the plate is square, so the collection efficiency is not a function of radius alone. Beyond the inscribed circle ($r > 125\,\mathrm{mm}$) only the four corner wedges of the plate are populated, so a fixed-width radial bin would mix a shrinking range of directions and its radial profile $\varepsilon(r)$ would reflect the geometry of the square rather than the detector. The uniformity is therefore quantified directly in the plate plane (x, y).

Two complementary uniformity measures. The whole-slab scan gives the collection efficiency as a *field* $\varepsilon(x, y)$, the position-binned image of Eq. 3. We characterise this field in two ways. (i) The *normalised map* $\varepsilon/\langle\varepsilon\rangle$, with $\langle\varepsilon\rangle$ the plate-averaged efficiency, is shown for each layout in Figs. 25–28 in 5 mm cells. It is a dimensionless field of unit mean that shows *where* the response is high or low and the spatial pattern of the asymmetry (for instance a top–bottom gradient), but *not* the absolute efficiency. (ii) The *two-dimensional contrast*

$$\kappa_{2\mathrm{D}} \;=\; \frac{\max_{\mathrm{cell}}\varepsilon}{\min_{\mathrm{cell}}\varepsilon}$$

collapses the field to a single number, the ratio of the brightest to the darkest region; it answers "how large is the spread" and is the quantity that ranks the layouts and reflectors. $\kappa_{2\mathrm{D}}$ is

evaluated on a coarser 5×5 grid of 50 mm cells, because on the 5 mm cells of the maps the per-cell statistics are too thin for a stable max/min—Poisson noise in the extreme cells would inflate it. The two measures therefore use different binning: the contrast quoted below is deliberately the *robust* dynamic range, and is smaller than the raw peak-to-trough range of the fine maps. No symmetry is assumed, so κ_{2D} is meaningful for the single-sided layouts 3 and 4 as well as for the four-sided ones.

The direction-sensitive contrast κ_{2D} ranks the reflectors consistently in *every* layout,

$$\mathrm{TiO_2} \;>\; \text{bare} \;>\; \mathrm{Al} \;\gtrsim\; \text{glossy},$$

i.e. the diffuse TiO_2 paint is the *least* uniform wrapper and glossy paper the most uniform. The diffuse paint is also the dimmest direct-coupled wrapper, while the bright specular wrappers are the most uniform: the specular transport mixes the light so thoroughly that the entry position is largely erased ($\kappa_{2D} \approx 1.1$–1.3), whereas the diffuse paint retains a pronounced positional response. The contact-limited bare plate sits between them because the holder loss is itself position dependent. Numerically the TiO_2 contrast grows from $\kappa_{2D} = 1.911$ and 2.579 in the four-sided layouts 1 and 2 to 8.146 and 8.311 in the single-sided layouts 3 and 4; the bare plate spans 1.36–3.82, aluminium 1.08–1.28 and glossy 1.07–1.28. A muon crossing anywhere on the plate is therefore collected within about 30 % of the most favourable point for Al/glossy, within a factor of 3.8 for the contact-limited bare plate, and within a factor of 8.3 for TiO_2 in the single-sided layouts. Figure 24 collects all sixteen values as bar charts, one panel per layout; Table 9 lists the same values numerically, resolving the near-unity entries that the linear bar scale cannot.

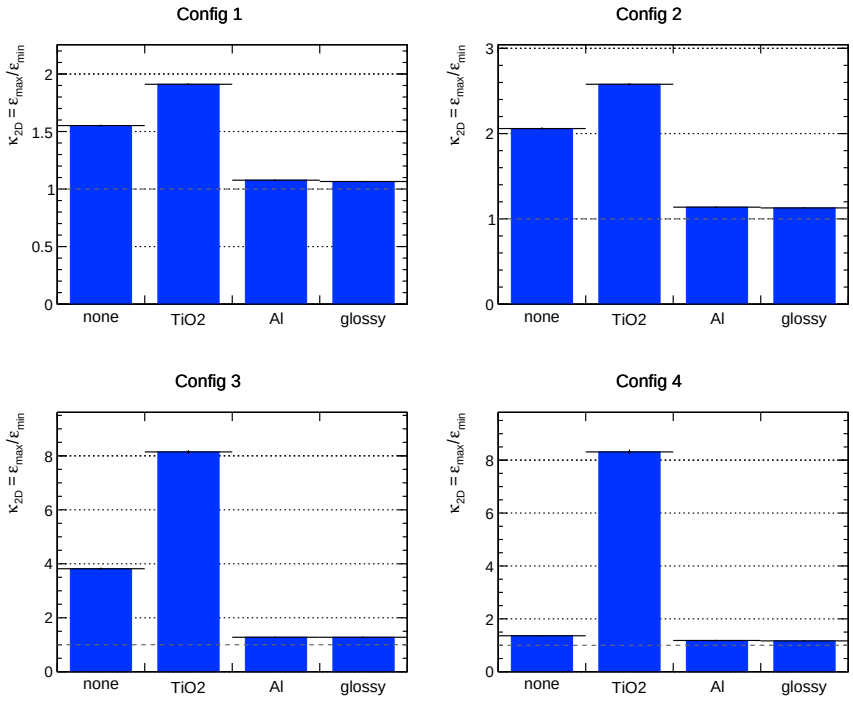


Figure 24: Full-plate position contrast $\kappa_{2D} = \varepsilon_{max}/\varepsilon_{min}$ over the 50 mm cells of the whole-slab map, one panel per layout and one bar per reflector (each panel on its own linear scale). This is the same kind of comparison as the three near-edge patches, but evaluated over the entire plate; the dashed line marks unity (a perfectly uniform response). TiO_2 shows by far the largest contrast in every layout, the contact-limited bare plate is second, and Al and glossy paper stay closest to unity.

Table 9: Full-plate position contrast κ_{2D} over the 50 mm cells of the whole-slab map, per layout and reflector, with the propagated Poisson uncertainty. These are the values plotted in Fig. 24; the table resolves the near-unity entries that the linear bar scale cannot, and shows the TiO_2 > bare > Al $\gtrsim$ glossy ordering in every layout. The number in parentheses is the uncertainty in the last digits of the preceding value.

Reflector	Config 1	Config 2	Config 3	Config 4
Bare	1.549(5)	2.060(9)	3.821(49)	1.359(9)
TiO_2	1.911(7)	2.579(12)	8.146(76)	8.311(92)
Al	1.078(3)	1.138(5)	1.283(6)	1.179(7)
Glossy	1.067(2)	1.131(3)	1.276(5)	1.166(6)

The single-sided layouts are the most demanding. The largest TiO_2 two-dimensional contrast of the entire study appears in config 4 ($\kappa_{2D} = 8.311$), just above config 3 (8.146): the front–back asymmetry of the diffusely wrapped plate is far stronger than any radial average reveals. The Al and glossy wrappers in the same layouts stay below 1.3, whereas the contact-limited bare plate reaches 3.82 in config 3. A single-sided layout is simply not described by a radius, so only the full (x, y) map—not a radial profile—carries the information.

Figures 25–28 show the same response in normalised form, $\varepsilon/\langle\varepsilon\rangle$, for each of the four layouts, on a shared *logarithmic* colour scale. A single linear scale cannot hold the range: for the diffuse TiO_2 paint the response spans about a factor of 1.9–2.6 about its mean in the four-sided layouts and up to a factor of 8.3 in the single-sided ones, dimmest at the centre and brightest toward the edges and corners. Aluminium and glossy paper occupy a narrow band around unity, and the contact-limited bare plate lies between them. The two four-sided layouts (1 and 2) are nearly azimuthally symmetric, whereas in the single-sided layouts (3 and 4) the TiO_2 panel shows the one-sided read-out directly, the region opposite the sensors darkening.

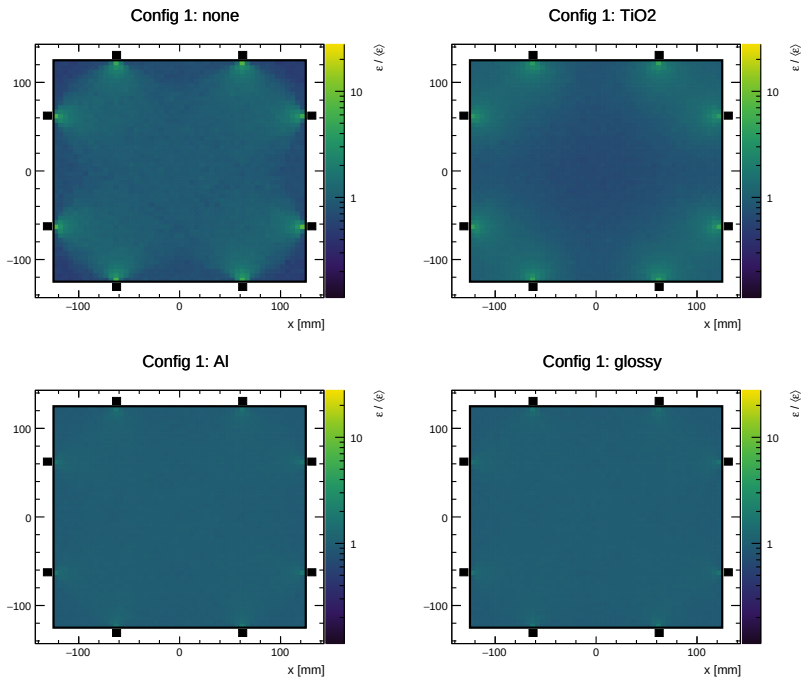


Figure 25: Whole-slab efficiency normalised to its plate average, $\varepsilon/\langle\varepsilon\rangle$, for config 1, one panel per reflector (5 mm cells, the same binning as Figs. 20–23, on a shared *logarithmic* colour scale). The slab outline and the SiPM positions (black squares), or for config 4 the WLS fibre clamp (red bar) with its read-out SiPM, are overlaid as in Fig. 23. The TiO_2 panel spans a factor 1.91 about its mean, dimmest at the centre and brightest at the edges and corners; Al and glossy occupy a narrow band around unity, while the contact-limited bare plate shows a modest inner deficit.

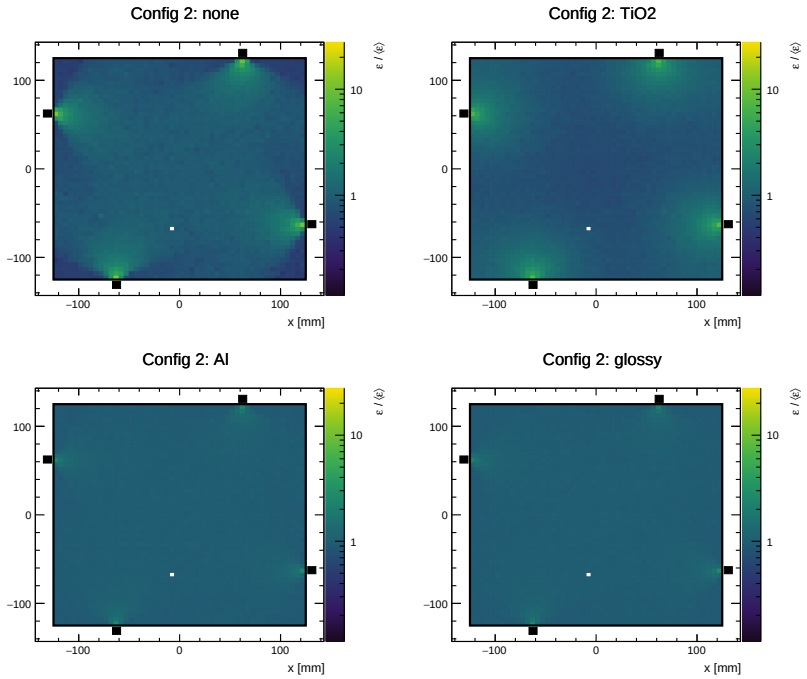


Figure 26: As Fig. 25 but for config 2 (four SiPMs, one per lateral face). The patterns are almost indistinguishable from config 1—the four-sided read-out is effectively azimuthally symmetric—with only the overall normalisation (hence the absolute efficiency) reduced.

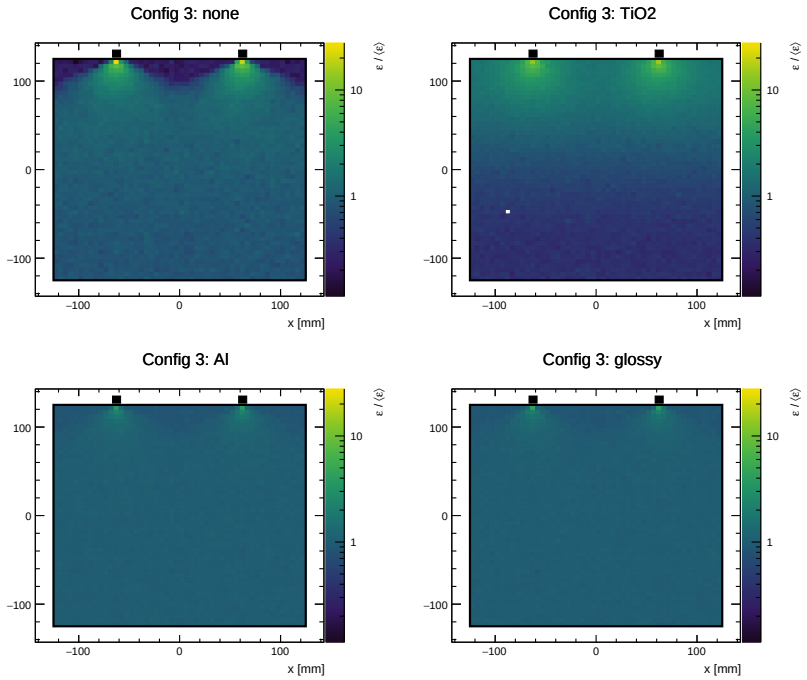


Figure 27: As Fig. 25 but for config 3 (two SiPMs, both on the top $+y$ face). The one-sided read-out breaks the azimuthal symmetry: the TiO_2 panel shows a pronounced top–bottom gradient, with the response falling toward the $(-y)$ face opposite the sensors. Al and glossy paper remain nearly flat, while the contact-limited bare plate shows a weaker version of the same asymmetry.

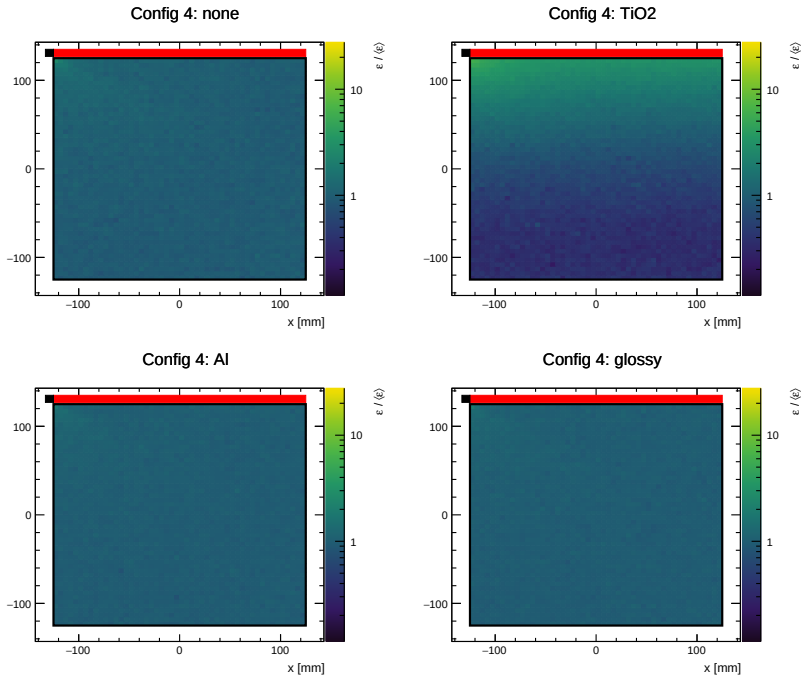


Figure 28: As Fig. 25 but for config 4 (single SiPM + WLS fibre clamp along the top $+y$ edge, read out at the top-left corner). The single fibre read-out gives the TiO_2 panel a very strong asymmetry, the response falling from the middle of the plate toward the $(-y)$ side opposite the sensors; it is essentially as strong as in config 3 ($\kappa_{2D} = 8.311$ versus 8.146), just above it.

5.4 Arrival-time structure and pulse persistence

The collection-efficiency maps above treat every detected photoelectron as a point; the new per-photoelectron timing columns of the `sipmHits` tree let the *arrival-time structure* of the detected light be examined as well. Figure 29 shows the oscilloscope-style persistence map of the arrival-time profile for the four layouts (glossy paper). For each event the per-SiPM arrival times are summed over the layout's sensors and histogrammed in time, giving a trace of photoelectron count versus arrival time; stacking events and recording, for every 0.1 ns time bin, how many events showed a given count, yields a two-dimensional map whose abscissa is the arrival time, whose ordinate is the instantaneous photoelectron count, and whose colour is the number of events. The rising edge and the decaying tail therefore show how the detected light builds up and dies away, and the vertical spread at a given time is the event-to-event fluctuation of the instantaneous rate. For the three directly-read layouts the modal mean arrival times are 3.45 ns, 3.65 ns and 3.55 ns, and the prompt peaks reach 62, 63 and 47 photoelectrons per 0.1 ns bin in configs 1–3; the single-fibre WLS read-out (config 4) peaks markedly later, at 9.45 ns and at only 12 photoelectrons per bin. All four panels share a single linear amplitude axis of 0–30 p.e. at 1 p.e. per bin; the very top of the prompt peak in configs 1–3 lies above this range and is off-scale. The fractions arriving within 5 ns are 31.6, 30.3, 29.7 and 7.3 % for configs 1–4, within 10 ns 63.2, 61.2, 60.2 and 32.8 %, while the late ($t > 20$ ns) tail carries 10.8, 12.3, 13.1 and 24.8 % of the photoelectrons.

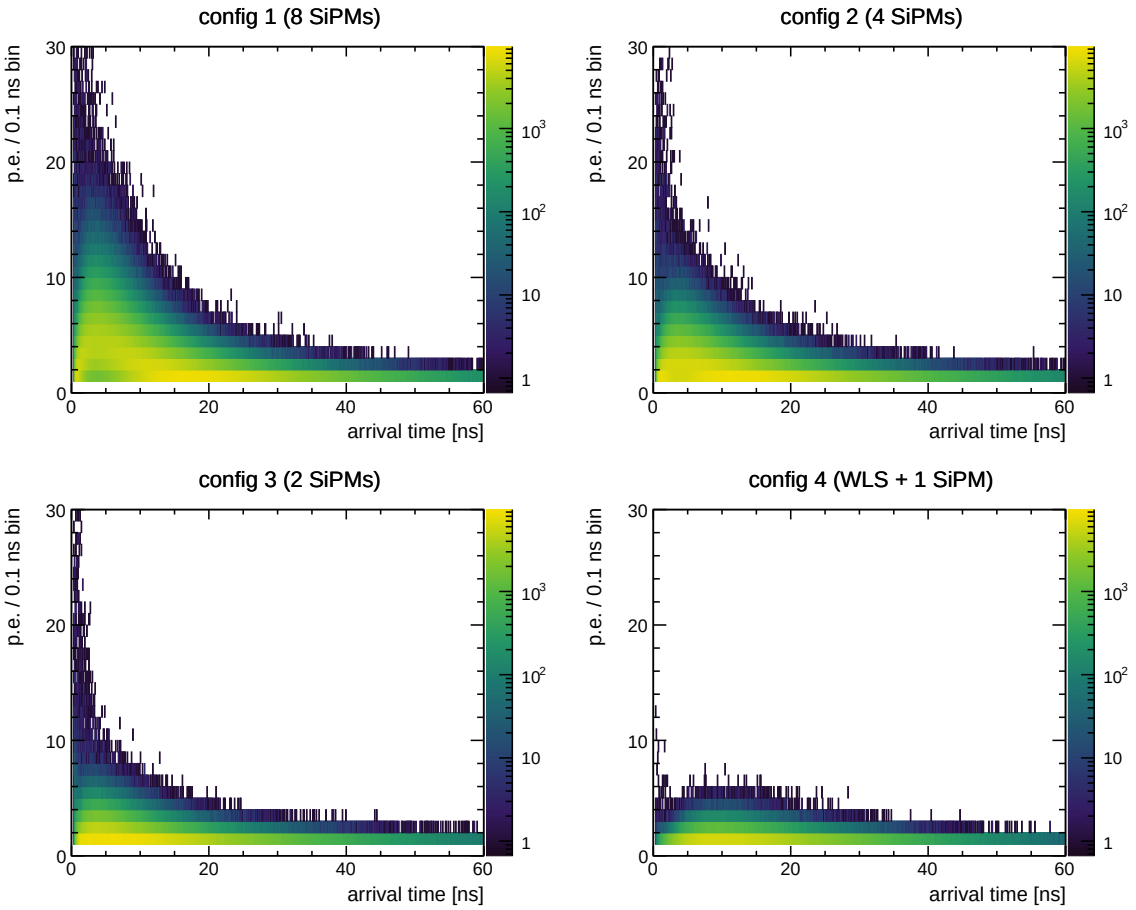


Figure 29: Arrival-time persistence map of the detected photoelectrons, one panel per layout (glossy paper, whole-plate irradiation). The abscissa is the global arrival time, the ordinate the number of photoelectrons in the corresponding 0.1 ns time bin (summed over the layout's SiPMs), and the colour the number of events reaching that point (shared logarithmic scale). Empty time bins are omitted; the ordinate is shared across the four panels and linear over 0–30 photoelectrons per bin (1 p.e. per bin), the arrival-time axis over 0–60 ns. The prompt rise reflects the direct scintillation and Čerenkov light, while the slower tail is the multi-bounce transport and delayed (self-absorbed and re-emitted) light. The WLS layout (config 4) builds up roughly 6.0 ns later than the others, its prompt peak shifted by the fibre absorption and re-emission.

The same arrival-time data can be collapsed into one average waveform per layout by averaging the per-event traces of Fig. 29 over events; Fig. 30 overlays the four averages. This makes the amplitude hierarchy and the time shifts explicit. Integrating the mean rate over the full 60 ns window recovers 518.8, 274.0, 140.2 and 67.5 photoelectrons per event for configs 1–4 (the same ordering found in the collection-efficiency maps), and the rising edges of the three directly-read layouts coincide within a fraction of a nanosecond, whereas the WLS read-out of config 4 rises about 6 ns later and decays more slowly – its late tail stands out on the logarithmic ordinate as a straighter, shallower fall.

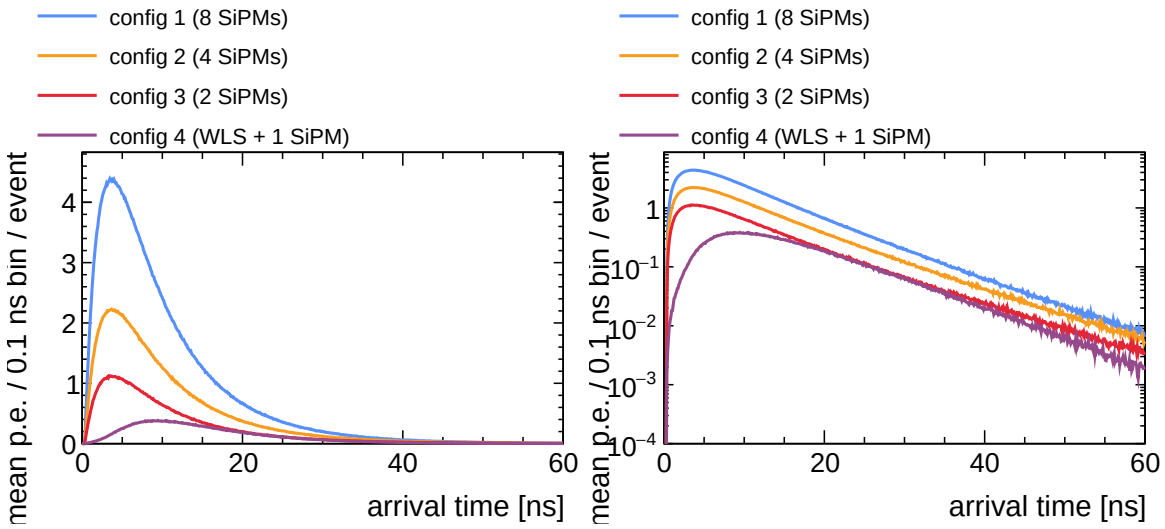


Figure 30: Average arrival-time waveform of the detected photoelectrons, one curve per layout (glossy paper, whole-plate irradiation), obtained by averaging the per-event traces of Fig. 29 over events. The abscissa is the global arrival time and the ordinate the mean number of photoelectrons in each 0.1 ns bin per event. The left panel uses a linear ordinate and the right a logarithmic one; all four configurations are drawn on each panel (the prompt peaks of configs 1–3 nearly coincide and extend above the linear range). The integrated mean yields are 518.8, 274.0, 140.2 and 67.5 photoelectrons per event for configs 1–4. The WLS layout (config 4) peaks roughly 6.0 ns later than the others and retains a long, slowly-decaying tail, the average-waveform counterpart of the persistence map.

Figure 31 combines the two offline observables: the arrival time and the wavelength of each detected photoelectron. Configurations 1–3 collect the blue scintillation spectrum directly, with a photoelectron-weighted mean of 456.7–457.2 nm, whereas the Y-11 WLS read-out of config 4 is green-shifted to 509.5 nm, consistent with the steady-state spectra of Fig. 14. In every layout the spectrum reddens with arrival time: the mean wavelength rises from 450.9 nm ($t < 5$ ns) to 467.7 nm ($t > 20$ ns) for configs 1–3, and from 499.1 nm to 511.3 nm for config 4, while the rms spread narrows (37.9 $\rightarrow$ 35.5 nm and 35.8 $\rightarrow$ 22.6 nm, respectively). The depleted blue edge of the late light is the signature of self-absorption along the multi-bounce path and, in config 4, of the additional wavelength-dependent attenuation of the WLS fibres.

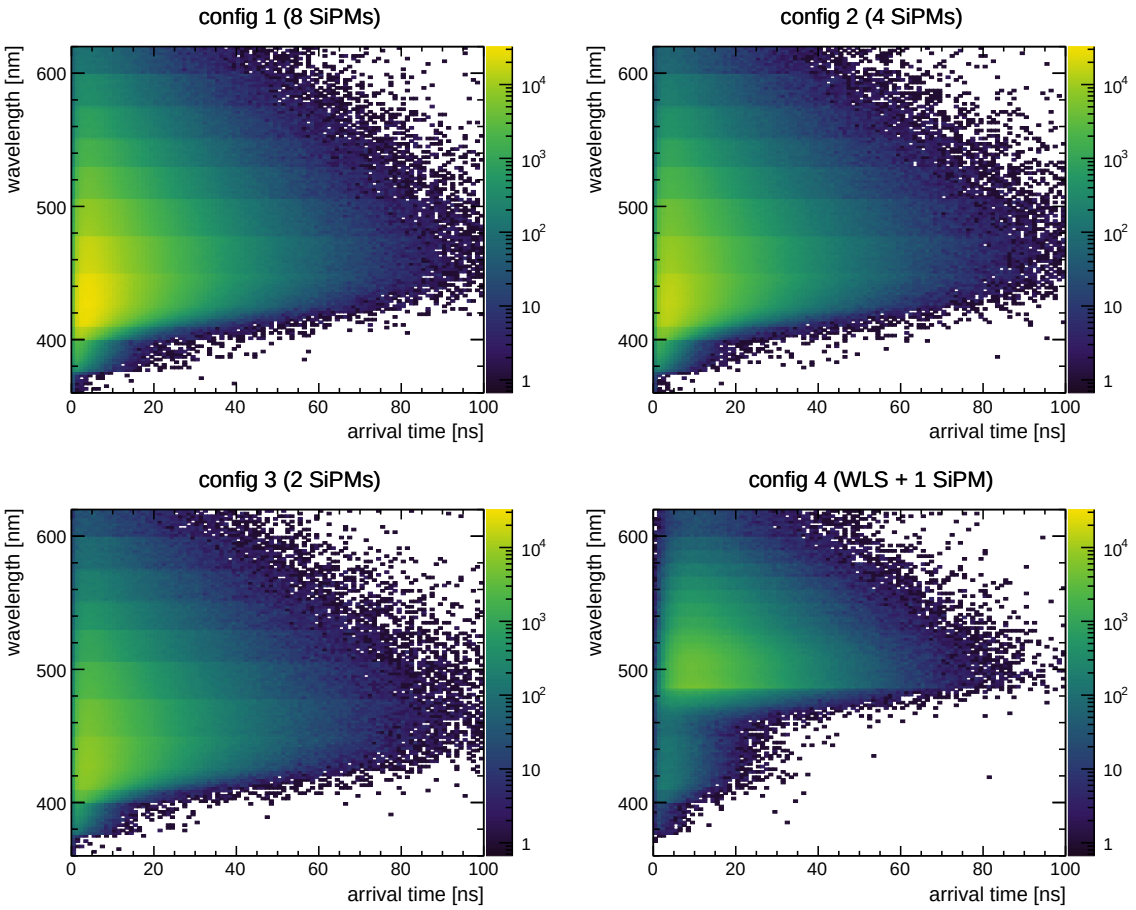


Figure 31: Detected wavelength versus arrival time, one panel per layout (glossy paper, whole-plate irradiation), on a shared logarithmic colour scale (number of photoelectrons). Configurations 1–3 populate a narrow band around the 457 nm blue scintillation spectrum; config 4 is green-shifted by the Y-11 WLS fibres. The time dependence of the mean wavelength traces the self-absorption of the shorter-wavelength light over the multi-bounce path (from 450.9 nm to 467.7 nm in configs 1–3 and from 499.1 nm to 511.3 nm in config 4, between the early and late arrival windows).

6 Caveats and open items

Optical-track stall (resolved). During earlier matrix production runs a rare stall was observed: a single worker thread would enter a tight loop (100% CPU) while the remaining threads waited at the event-level barrier, freezing the run. It was traced to insufficiently robust photon tracking in the *tessellated* config-4 fibre solid, where near-coincident facet boundaries could leave a track stuck. The production config-4 geometry now uses the analytic, mitered `G4CutTubs` fibre legs (Sec. 2.4) instead, whose straight, planar miter interfaces eliminate that class of degenerate boundary; the stall has not recurred since. As a safety net the production workflow still wraps every task in a timeout with automatic retry.

Bare-holder contact is effective, not measured. Modelled with a real 1.2 mm air gap between the polished plate and the black holder ($f = 0$), the bare configuration is a near-lossless TIR light guide and is the *second*-brightest wrapper in every layout, not the weakest. Reproducing the experimental suppression requires the plate–holder interface to be optically *lossy*, i.e. partial real contact (frustrated TIR). The extended contact scan shows that the crossing fraction is ≈ 0.017 at the centre in every layout, 0.003–0.009 at the edge/corner in configs 1–2, and 0.039–0.041 at the edge/corner of config 3. No single f matches every cell exactly; the production nominal $f = 0.045$ is the smallest sampled fraction that places all nine cells below TiO_2. The

fraction is an effective area fraction, not a measured holder gap, so $f = 0$ is retained as the clean-air reference and lower bound on the contact loss.

Position trend in TiO_2. For the air-backed Al and glossy wrappers the response is nearly flat across the three patches (Al 1.5 % to 3.4 %, glossy 0.8 % to 3.2 %). The contact-limited bare plate varies more (4.7 % to 18.9 %) because the holder loss is itself position dependent. The diffuse TiO_2 paint is the only wrapper with a clear position dependence, and its sign depends on the layout: in the four-sided layouts 1 and 2 it shows corner > edge > centre (a factor $\approx$1.75), whereas in the single-sided layouts 3 and 4 it shows centre > edge $\approx$ corner. Since `groundfrontpainted` is purely Lambertian (no TIR, no specular reflection), the sign follows the geometry of the nearest painted walls relative to the illuminated sensors. No corresponding trend is reported in Ref. [1] and it should be confirmed with independent simulation settings before being interpreted physically.

Scale and normalisation. The absolute efficiencies are model-dependent: they are fixed by the assumed PDE curve, the grease coupling, and the absorbed fraction of the tabulated reflectivities. The SiPM breakdown voltage and gain (which drive the PDE) are also temperature-dependent [9], an effect not modelled here. The robust results are the *ordering* (glossy paper brightest and most uniform, air-backed aluminium second, contact-limited bare plate weakest in the direct-coupled layouts, and TiO_2 weakest in the WLS layout) and the full-plate contrast hierarchy, not the absolute efficiencies.

7 Summary

A Geant4 optical simulation of a 250 mm×250 mm×10 mm polished plastic scintillator read out by side-mounted SiPMs has been developed and applied to a full matrix of four SiPM layouts and four reflector treatments at three beam positions (120 000 events), complemented by a uniform whole-plate irradiation (400 000 events), a 99 000-event bare-contact scan and an 18 000-event aluminium-backing scan (637 000 events in total). The optical physics—unified surface model, finish selection, photocathode PDE via the boundary process, and per-event photon accounting—has been verified against the Geant4 source.

With the calibrated surfaces the direct-coupled layouts reproduce the experimentally observed ordering: glossy paper is the brightest and most uniform wrapper, air-backed aluminium reaches about 92 % of its yield, the contact-limited bare plate is the weakest, and the diffuse TiO_2 paint is the most position dependent. In the WLS layout the ordering is instead TiO_2 < bare < Al < glossy, because the black holder is absent there. The air-gap model ($f = 0$) remains the clean reference in which the bare plate is second only to glossy paper; the production contact fraction $f = 0.045$ is the smallest sampled value that places all nine direct-coupled cells below TiO_2, at the cost of over-suppressing the four-sided centre. The aluminium–grease branch is retained as the lower-bound systematic: air-backed aluminium is consistent with the measured near-equality of the two best wrappers, while grease coupling would put the foil below the bare plate.

Keeping the three irradiation patches separate—rather than collapsing them to a single beam-axis radius—shows that the air-backed Al and glossy responses are nearly flat (position contrast 1.5 % to 3.4 % and 0.8 % to 3.2 %), whereas the contact-limited bare plate varies by 4.7 % to 18.9 % and the diffuse TiO_2 paint by 74.8 % to 87.4 %. Extending the scan to the whole plate sharpens this picture: the full-plate contrast is largest for TiO_2 (κ_{2D} = 1.91–8.31), reaches 1.36–3.82 for the bare plate, and stays at or below 1.28 for Al and glossy. The plate-averaged efficiency scales roughly with the number of directly coupled SiPMs, while the single-SiPM WLS layout is comparable to the two-SiPM config 3. Its arrival-time distribution is delayed by about 6 ns and its spectrum is green-shifted to 509.5 nm by the Y-11 fibres, confirming that the WLS path is active. Absolute efficiencies remain model-dependent (PDE, coupling, and tabulated reflectivities); the robust results are the wrapper and layout orderings and the full-plate contrast hierarchy.

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